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The economics of skyscrapers: A synthesis[☆]

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ABSTRACT

We document that skyscraper growth since the end of the 19th century has been driven by a reduction in the cost of height, increasing urbanization, and rising incomes. These stylized facts guide us in developing a competitive open-city general equilibrium model of vertical and horizontal city structure. We use the model to show that (i) vertical costs and benefits affect the horizontal land use pattern within cities; (ii) the causal relationship between skyscrapers and urbanization is bi-directional; and (iii) height limits reduce the size of large cities, leading to lower agglomeration economies, productivity, and urban GDP. We substantiate the model's predictions by novel estimates of urban height gradients.

1. Introduction

The study of the *horizontal* structure of urban space has been one of the main research areas in urban economics ever since the birth of the field (Alonso, 1964; Mills, 1967; Muth, 1969). While iconic skylines have long become distinctive features of cities over the course of the 20th century, it is not until recently that urban economics research into the *vertical* structure of cities has gained momentum.

We bring together recent research on costs (Ahlfeldt and McMillen, 2018; Barr, 2010) and benefits (Barr, 2016; Liu et al., 2018; 2020) of building height, which we integrate into a general equilibrium model of the vertical and horizontal structure of an open city. We use the model to engage with the following questions concerning the positive and normative economics of skyscrapers: How do vertical costs and benefits shape the internal structure of cities? How does the distribution of economic activity between cities and rural hinterlands affect the vertical size of cities and vice versa? What are the consequences of distortionary policies that constrain the vertical size of cities on the spatial distribution of economic activity and welfare? We complement our theoretical analyses with novel evidence on urban height gradients to guide our modeling choices and confirm the model's predictions. From our theoretical and empirical analyses, as well as our reading of the literature, we conclude that vertical costs and benefits, horizontal land use patterns, urban growth, and welfare are all mutually dependent in ways that leave plenty of room for future research.

In developing a model that integrates vertical and horizontal spatial structure, we must engage with the obvious question of how skyscrapers emerge.[1] Therefore, we start by exploiting a unique data set that blends building-level data from commercial and non-profit providers covering the entire planet and hand-collected data for selected cities spanning 150 years (see Online Appendix Section A for details) to produce a battery of stylized facts that speak to the determinants of building height. We show that since 1900, there were waves of skyscraper development *when* there were innovations in construction technology and *where* there was economic growth. With few exceptions, the tallest skyscrapers today are not outliers within the skylines of their host cities. Even the Empire State Building, which ruled the height ranking for longer than any other building, delivered a decent return on investment over its lifetime. Our conclusion is that the canonical approach that assumes profit-maximizing developers in a competitive market provides a reasonable description of height decisions, even for most tall buildings.

[1] Canonical urban models assume profit-maximizing developers in a competitive market (Brueckner, 1987; Duranton and Puga, 2015; Ahlfeldt et al., 2015). However, if developers compete to win the prize of being the tallest, heights may not be justifiable by economic fundamentals (Helsley and Strange, 2008; Barr, 2010; 2013; Barr and Luo, 2020).

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With this in mind, we have developed a new general equilibrium model of vertical and horizontal city structure to study the causes and effects of vertical growth. To engage with our main research questions, we design the model to serve three purposes. First, the model should rationalize the remarkable differences in building heights *within* cities that are characteristic of metropolitan skylines and account for how use-specific vertical costs and benefits affect the horizontal land use pattern. Second, the model should address the mechanisms that drive the close relationship between urbanization, vertical growth, and rising incomes *between cities*, which we document in our data. Third, the model should be suitable for the evaluation of a common form of land use regulation that constraints vertical space: building height limits. Here, we are interested in the effects that this distortionary policy has on the spatial allocation of economic activity *within* and *between* cities.

Our basic set up is an open-city model (Roback, 1982) with endogenous land use (Duranton and Puga, 2015) and returns to agglomeration in the form of higher productivity. We borrow the supply side of land markets from Ahlfeldt and McMillen (2018), i.e., we impose that the marginal cost of supplying one unit of floor space increases in height at a use-specific elasticity (Barr, 2010). Thus, the model features a labor market related agglomeration force in the form of higher wages and a housing market related dispersion force in the form of higher rents.[2] As for the geography, we consider a linear city where residential and production amenities differ across both horizontal and vertical space. We summarize accessibility benefits by a monotonic function of distance from a historic core and allow for greater production and residential amenity values at higher floors to rationalize recent evidence on positive vertical rent gradients (Liu et al., 2018; Danton and Himbert, 2018).

Perfect competition among firms and perfect mobility of residents leads to perfect spatial arbitrage so that differences in amenities map into differences in rents in horizontal and vertical space. Profit-maximizing developers respond to higher rents by building taller. Perfect competition results in full capitalization of developer profits in land rents. Land is allocated to the use that generates the highest land rent, pinning down the endogenous size of the central business district (CBD) and the residential zone. In equilibrium, the city-specific wage and population, as well as location-specific floor space rents adjust such that labor and land markets clear and the utility within the city equates to the exogenous reservation utility. In this setting, an exogenous change in any parameter will generally lead to adjustments in all endogenous outcomes. Hence, the model is well suited to account for the mutual dependence of use-specific *vertical* costs and benefits, *horizontal* patterns of land use, and the size and productivity of a city.

Before we can use the model to quantitatively evaluate these mutual dependencies, we need to settle on canonical values for the model's structural parameters. Two of them deserve particular empirical attention because evidence is scarce: The height elasticity of per-unit construction cost and the height elasticity of per-unit floor space rent, which impact the costs of and returns to height. For the former, we estimate a range from 0.1 for smaller structures to 0.25 for tall commercial and 0.56 for tall residential structures. For the latter, we estimate a value of about 0.07 for residential units in New York City, as well as in Chicago. For New York City, we estimate a value of 0.033 for commercial rents. These estimates reveal that the costs of, as well as the returns to, height are greater for residential than for commercial land use.[3]

The importance of accounting for such use-specific differences to understand the urban height gradient becomes apparent in the first application of our model. Under the baseline parametrization, the model delivers canonical height and rent gradients that are downward-sloping in distance from the city center and a central business district (CBD) surrounded by a residential zone (Duranton and Puga, 2015). The lower net-cost of height for commercial buildings results in steeper commercial height and land rent gradients. Importantly, we obtain the novel prediction that the commercial and residential land bid rents intersect nearer to the center than the floor space rent gradients. This pushes the CBD boundaries inwards and generates discontinuities in the height and the floor space gradients at the edge of the CBD. Hence, our model rationalizes why tall CBDs often stand out relative to the surrounding residential zones through endogenous mechanisms and without requiring exogenous zoning or spatial discontinuities in the distribution of amenities.

The model predictions for the height gradient are consistent with patterns that we find in the data. We estimate an elasticity of building height with respect to distance from a "prime location" (business centers identified by Ahlfeldt et al., 2020) of about -0.25 for a global set of cities in 1900 and 2015. It appears that demand-side forces pushing for a flatter height gradient and supply-side forces pushing for a steeper height gradient have largely offset each other in their effect on the slope of the urban height gradient over the 20th century. Our estimates confirm that commercial rent gradients are steeper than their residential counterparts. We also find evidence for a discontinuity in the height gradient at the edge of CBDs. Conditional on smooth gradients inside and outside the boundary, there is a jump in building heights of slightly more than 20% as one steps into the CBDs of the U.S. cities in our data. Finally, we discretize space and solve for the distributions of residential and commercial amenity under which the model matches the land use pattern and the skyline of Chicago. We find that small differences in the amenity value are able to rationalize a fuzzy height gradient with large differences in building heights. Under the inverted amenity values, the model generates a distribution of land values that closely resembles observed data, lending support to the parametrization of our model.

In the second application of the model, we turn to the obvious stylized fact that skyscrapers are a big-city phenomenon. A standard prediction of urban land use models is that, because the horizontal expansion is constrained by commuting costs, a larger city size maps into greater structural density, i.e., taller buildings (Brueckner, 1987; Duranton and Puga, 2015). A separate literature advocates the reverse causality by arguing that a favorable geography (solid bedrock near the surface) facilitates the construction of tall buildings, leading to agglomeration and greater productivity (Rosenthal and Strange, 2008; Combes et al., 2011). In a series of comparative statics that engage with major long-run trends observed over the 20th century, we illustrate how the causal relationship between building height and city population is, indeed, bi-directional. A rise in returns to agglomeration increases the wage and attracts workers to large cities, causing skyscrapers to emerge via a demand-side channel. Similarly, a reduction in the horizontal cost of travel not only leads to a horizontal expansion, but also to a vertical expansion, since the heightened attractiveness of the city results in greater demand for floor space. In contrast, a reduction in the cost of height leads to taller buildings via a supply-side channel and lower floor space rents, which causes the city to grow until the equilibrium is restored. Hence, the sizable reduction in the cost of building tall could be an important driving force of urbanization that has gone unnoticed in a literature concerned with the growth of cities.[4]

In the third application of the model, we evaluate how the spatial structure of cities responds to height limits. In keeping with intuition, a vertical compression leads to a horizontal expansion of the CBD. The effect on the residential zone, however, is the opposite. In our open-city model, the supply-driven increase in floor space rents leads to out-migration until an equilibrium is restored at a smaller population living on less horizontal space. The wage level falls owing to smaller agglomeration economies and, eventually, rents fall below the free-market equilibrium to restore the reservation utility. Our model predictions substan-

[2] See for a summary of the related empirical evidence on the economic effects of density, see Ahlfeldt and Pietrostefani (2019).

[3] We also find that costs and benefits likely decreased over time.

[4] See, for example, Glaeser et al. (1992); Henderson et al. (1995); Duranton and Turner (2012).

tiate a popular notion in the literature that height regulation can have large and negative welfare consequences (Glaeser et al., 2005); yet, they also reveal that, if workers are mobile, relaxing height restrictions in the most expensive cities may have smaller impacts on affordability than advocates may wish for (Glaeser et al., 2005), at least in the long-run.[5]

The remainder of the paper is organized as follows. Section 2 introduces our data and provides stylized facts that motivate some of the choices that we made when developing our model in Section 3. Section 4 discusses our choices of parameter values. Section 5 compares the model predictions for vertical and horizontal spatial structure with the evidence. Section 6 uses the model to explore the reciprocal relationship between urbanization and vertical growth. Section 7 analyzes the impacts of height caps on welfare and the vertical and horizontal urban structure. Section 8 concludes by summarizing some priority areas for future research. For the interested reader, we complement each section in this paper with a dedicated section in the online appendix in which we provide further theoretical and empirical analyses, discuss the related literature in more detail, and suggest directions for future research.

2. Data sets and stylized facts

Before we develop our model of vertical and horizontal structure in Section 3, it is useful to review some stylized facts that suggest that a conventional competitive model with an emphasis on profit-maximization is a reasonable description of height decisions. We find that vertical growth appears to be a rational response to changing demand and supply conditions. Suspiciously-tall buildings, whose height is likely determined by non-economic motives, appear to be the exception rather than the rule. We introduce the data set that we use in this and the subsequent sections in Section 2.1 before we turn our attention to the spatiotemporal diffusion of skyscrapers in Section 2.2 and the particularities of super tall buildings in Section 2.3.

2.1. Data

For our empirical work, we combine data that fall into four categories, which we briefly discuss below. More information on sources can be found in Online Appendix Section A. First, we exploit geo-coded data sets containing building heights, use, completion dates, and construction costs of tall buildings all over the world that are available from commercial and non-for-profit data providers, such as Emporis and the CTBUH Skyscraper Center. We complement these data with administrative data on all buildings for New York and Chicago obtained from the respective authorities.

Second, we use commercial rent and residential real estate prices with a geo-coded building reference for New York and Chicago from Cushman & Wakefield, Streateasy.com, Redfin.com, as well as geo-coded commercial rents for a wider set of global cities from SNL-S&P. For geo-coded land prices in Chicago and New York spanning more than a century, we rely on land values from Hoyt (1933), *Olcott's Land Values Blue Books of Chicago*, as well as vacant land sales from Ahlfeldt and McMillen (2018), Barr et al. (2018), Fred Smith (Davidson College), and Spengler (1930).

Third, we look at hand-collected data sets. We obtain gross and net floor areas for a global sample of buildings from Sev and Özgen (2009), Watts et al. (2007), Kim (2004), and Berger (1967). We collected cash flows for the Empire State Building throughout the 20th century from files located in the Hagley Museum Archives, newspaper articles, and SEC 10-K forms.

Fourth, we analyze global cross-sections of population, GDP, and national macroeconomic time-series that are readily accessible from institutions, such as the World Bank or MeasuringWorth.com.

2.2. Spatiotemporal patterns in skyscraper diffusion

Fig. 1 illustrates the height of the tallest skyscraper completed each year over the 20th and 21st centuries.[6] The pattern is cyclical, and there are several peaks throughout history. From 1908 to 1913, three buildings held the title of the tallest building in the world, all exceeding 200 m and all located in New York City. World War I induced a period of less ambitious construction before the next wave of skyscraper development culminated in the famous skyscraper race in which the Chrysler Building was defeated by the iconic 380-meter-tall Empire State Building (Bascomb, 2004).

Following the Great Depression and World War II, it took until the 1950s before the tallest buildings started to reach the levels of the late 1920s. A new benchmark was reached in the 1970s when the Twin Towers took the crown from the Empire State Building after nearly four decades. It was rapidly overtaken by the 440-meter-tall Willis (Sears) Tower in Chicago, the first record-breaking skyscraper outside of New York City. Following the oil crisis, there was a dip in tallest buildings until the 1990s when skyscraper development gained new momentum, this time in a much more international context. The Petronas Towers (1998) and the Taipei 101 (2004) pushed the limits to 452 and 508 m, respectively, before the Burj Khalifa set the record of 828 m in 2009. Since then, it has become the norm that the tallest completed structures within a year exceed 500 m (about 100 floors).

These waves of development during the late 19th century, the 1970s, and the turn of the 19th to the 20th century coincide with both periods of economic growth in North America and Asia and seminal innovations in construction technology.[7] Early skyscrapers were facilitated by the steel-framed skeletal structure and the electric elevator that become available before 1900. From the 1960s onward, mainframe computing allowed for the implementation of new structural techniques that require less steel per cubic meter, such as the framed-tube structure. Further refinements at the beginning of the 21st century, such as the buttressed core in the Burj Khalifa, coincide with yet another jump in the height of the tallest completions (see Online Appendix Section B.1 for a review of the technological history of skyscrapers).

The same cyclically is visible in the number of completed skyscrapers, depicted in Fig. 2. Driven by Asia, the pace of vertical growth has accelerated since the 1990s. In 2002, Asia took over from North America as the region with the largest cumulative number of skyscrapers. Today, 43% of the world's 150-meter or taller towers are in China alone (including Hong Kong).

To quantify the positive long-run trends in heights and completion counts of 150-meter or taller buildings, it is useful to regress the log of heights and completions against a yearly time trend. Over 120 years, the heights of the tallest building have increased at an average rate of 1.3%. The completions count has increased at an even larger percentage of 4.9%. Given that simple log-linear trends explain 63% and 82% of the variation in tallest heights and skyscraper counts over time, it seems fair to conclude that pronounced vertical growth, historically, has been the norm rather than the exception (see Table A1 in Online Appendix Section B.2).

In Fig. 3, we take a closer look at the spatial diffusion of skyscrapers. The first skyscraper outside the U.S. was completed in Brazil (the Altino Arantes Building) in the 1940s. Skyscrapers reached many of the larger

[5] But it would generate sizable economic benefits, such as higher average productivity due to greater exposure to external returns to scale (Hsieh and Moretti, 2019). In any case, a complete welfare appraisal of height regulations will have to account for the reductions in the presumed negative externalities from building height, such as increased shadows, reduced sunlight, increased traffic congestion, or lost historic charm

[6] For simplicity, we define a "skyscraper" as an occupiable structure that is at least 150 m tall.

[7] Barr et al. (2015) finds that GDP Granger-causes height, but that height does not predict GDP.

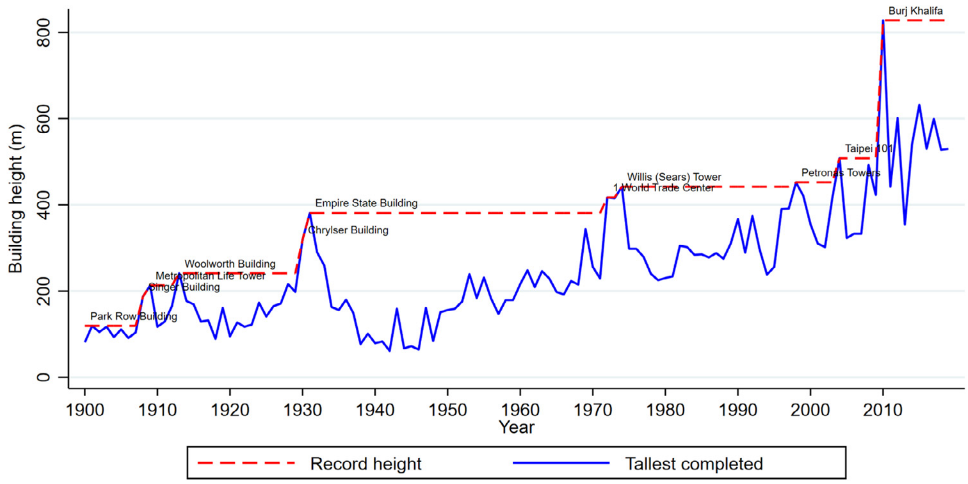


Fig. 1. Tallest completions Notes: Sources: Data from https://www.emporis.com/ and https://www.skyscrapercenter.com/.

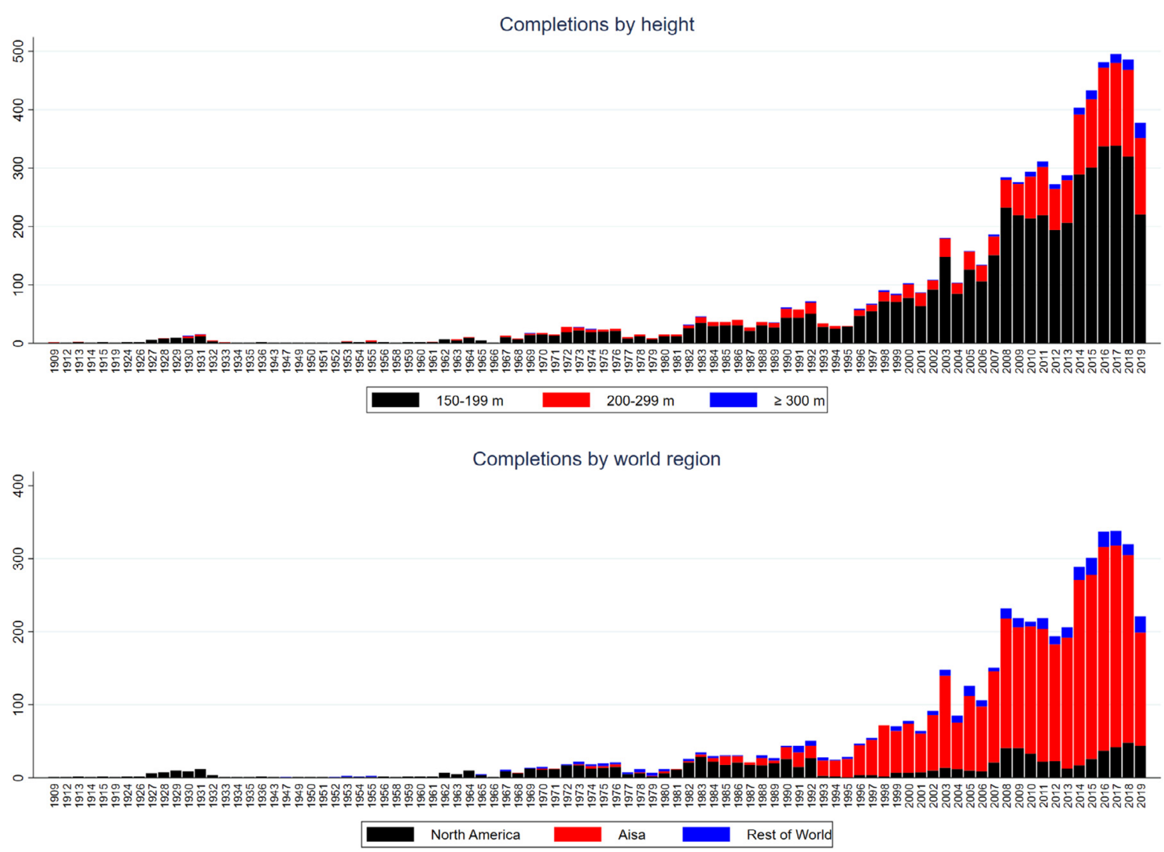


Fig. 2. Skyscraper completions by year Note: A skyscraper is a building that is at least 150 m tall. Source: https://www.skyscrapercenter.com/.

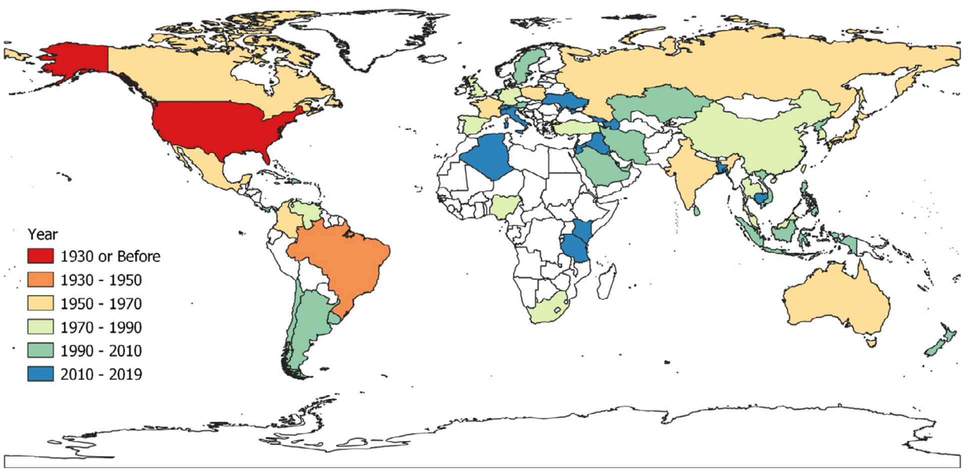
and economically more developed countries during the second half of the 20th century. Consistent with a shift in gravity from the developed to the developing world, economic growth has become the primary determinant of vertical growth, in addition to the level of GDP (see Figure A1 Online Appendix B.2).

Still, only 59 out of 193 nations (30%) have at least one skyscraper.[8] Some developed countries have restricted the adoption of the technology by means of land use regulation. For example, it took until the 2010s for skyscrapers to reach Italy and Switzerland. Ireland and Portugal do not have any skyscrapers (150 m or taller) to date.

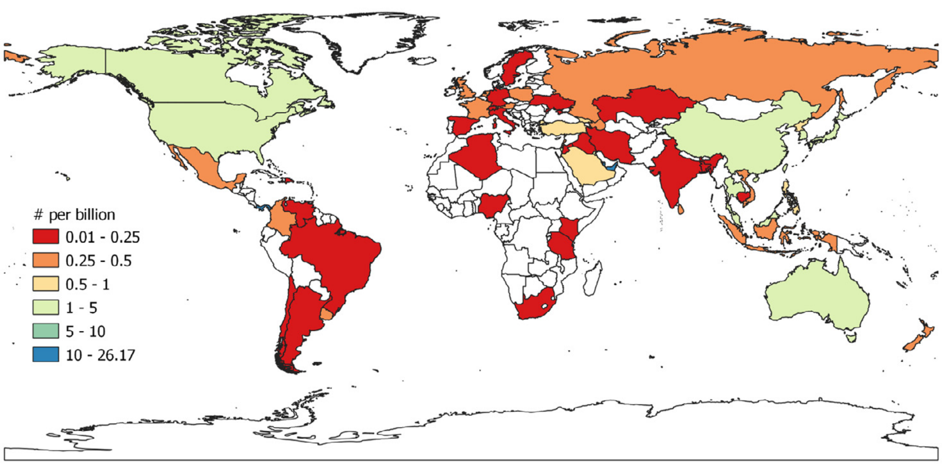
The degree of urban bias in the distribution of skyscrapers is striking. Small states in which the largest city dominates the city system, such as Panama or the United Arab Emirates, reach the highest penetration in per capita terms. 50% of the world's skyscrapers are located in just 17 cities, though a total of 315 cities worldwide have at least one. In terms of 150-meter or taller skyscrapers, Hong Kong leads the list with 353, with New York City coming in second at 281. Of the top twenty cities around the world, nine are in China (including Hong Kong). 16 out of 20 are in east Asia. Only three cities in North America (New York, Chicago, and Toronto) make the list.[9]

[8] 80% of all skyscrapers are in eight countries.

[9] See https://www.skyscrapercenter.com/, accessed May 1, 2020.



(a) Year of first skyscraper



(b) Skyscrapers per capita

Fig. 3. Skyscraperization by country Note: A skyscraper is a building that is at least 150 m tall. Source: https://www.skyscrapercenter.com/, as of 2018. Population data from https://data.worldbank.org/.

To summarize, it seems reasonable to conclude that skyscrapers tend to break height records when new technologies become available and where urbanization and economic growth create the necessary demand, unless regulation prevents the adoption of the technology.

2.3. "Too tall" buildings

While the spatiotemporal diffusion of skyscrapers appears to reflect the interplay of demand and supply conditions, there is nevertheless a common belief that the height of the tallest buildings is driven by non-pecuniary motivations. As an example, the "height race" in New York City during the 1920s led to the perception that developers of tall buildings, in aiming to dominate the skyline, seek to satisfy their egos rather than maximize profits (Barr, 2016).[10] Indeed, Helsley and Strange (2008) offer a game-theoretic model that rationalizes how developers vying to claim the prize of the "tallest building" build "too tall" to preempt rivals (see Online Appendix Section B.5 for a discussion of the related literature).

It is difficult to judge objectively by how much the height of an iconic building exceeds the fundamentally justified height. However, it is reasonable to expect that if pecuniary motives dominate, the second-tallest building in a city will be a good predictor of the height of the tallest building in a city. Indeed, we find a strong correlation across a large set of cities that adopted the skyscraper technology, as shown in Fig. 4, but there are some outliers, which include obvious candidates such as Dubai, home to the Burj Khalifa, the tallest building on the planet. If we turn to the heights of the second and third tallest buildings, however, the correlation is strikingly close.[11] Hence, it seems likely that economic fundamentals drive the heights of nearly all but a handful of the tallest buildings in a limited number of cities. Consistent with this interpretation, we find that only few buildings managed to hold the crown of being the tallest for a very long time. While the Empire State Building ruled height rankings for 40 years, the median duration is six years at the global and four years at the national scale (see Online Appendix Section B.4). This relatively rapid succession is consistent with a steady rise of the profit-maximizing building height, due to increasing demand and improving construction technology.

Closely related to the presumption that tall buildings serve non-pecuniary motives is the notion that they are not particularly profitable. For example, in 1930, Clark and Kingston (1930) wrote their book, *The Skyscraper: Study in the Economic Height of Modern Office Buildings*, to argue that those who called skyscrapers "freak buildings" misunderstood the economics of building tall. They argued that tall buildings were in-

[10] Bercea et al. (2005) argue that in the 16th century, the Roman Catholic church used cathedrals to signal their power when faced with competition from Protestantism.

[11] The only notable outlier is Ho Chi Minh City in Vietnam, where skyscrapers are a recent phenomenon.

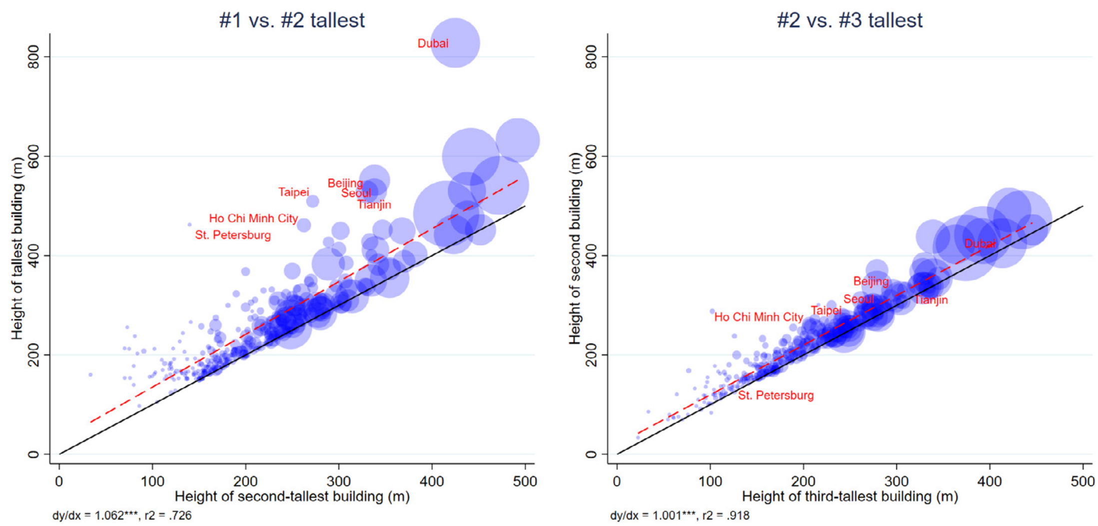


Fig. 4. Tallest building heights Note: Data covers 315 cities around the world with at least one skyscraper (heights ≥ 150 m. Marker size is proportionate to the number of skyscrapers in a city. Data are from Emporis (accessed in April, 2021).

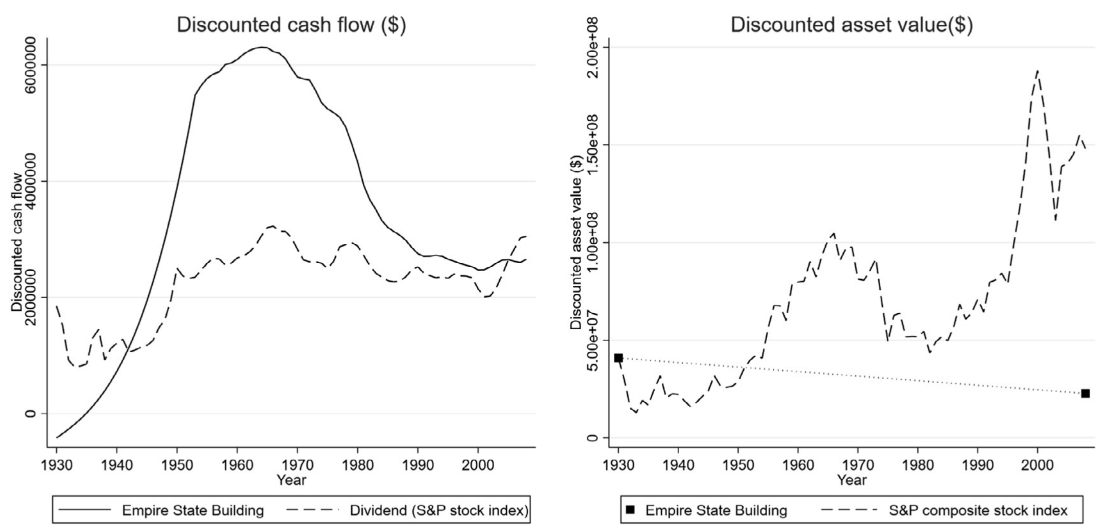


Fig. 5. Empire State Building vs. stock market Note: ESB asset value is total cost of structure and land in 1930 and land value in 2008 (as we assume that the structure has fully depreciated). For stock market, the asset value is the total investment inflated by the stock market index. Cash flow of the ESB is the net operating income (EBITDA). Cash flow of the stock market is the dividend (the product of the dividend yield and the asset value). All time series deflated by the cummulated opportunity cost of capital (risk-free rate). The net realized return for the ESB, at 5.4%, beats the stock market, at 4.3%. See Online Appendix Section B.4 for details.

herently a function of high land values, and they aimed to silence critics who felt that tall buildings were an inefficient use of land.

Critics, however, have not been convinced. The Empire State Building (ESB), in particular, became infamous as a stereotype of a loss-making "too tall" building (Kingwell, 2006). To move beyond anecdotes that mainly refer to high vacancy in the ESB during 1930s, we collected archival records and recent annual reports up until 2009, when a half billion dollar renovation marked the beginning of a new life for the ESB. Fig. 5 presents the key results of an ex-post financial case study whose details we report in Online Appendix Section B.4. Indeed, the ESB took a greater hit than the stock market during the Great Depression. Yet, the recovery was also stronger. During the 1950s, 60s, and 70s, the ESB's net operating income exceeds the dividend of a stock market portfolio with the same initial asset value as the ESB by a large margin.[12]

[12] During this period, the ESB was purchased by Harry Helmsley (in 1961) who built a reputation for developing a highly profitable portfolio that made him a real estate billionaire.

In terms of asset value, the stock market portfolio naturally outperforms the ESB as its structure capital, which accounts for about two thirds of the initial asset value, depreciates to zero. However, over the (first) life-cycle of the ESB, the higher discounted net operating income weighs more (as it occurs earlier). As a result, the ESB delivered a rate of return on investment of 5.4% on top of the risk-free rate, beating the stock market, which achieves 4.3%. Adjusting for the adverse economic conditions during the 1930s and 1940s, the return would have exceeded 8%, which was a good return for 1929 Manhattan (Clark and Kingston, 1930). The useful lesson for our model in the next section is that even the developers of the world's arguably most iconic skyscraper were not willing to compromise much on their ambition to make profits.

3. Model

Guided by the stylized facts introduced in Section 2, we now develop a general equilibrium model of urban land use. The model features various canonical elements of standard land use models reviewed by Duranton and Puga (2015). The novel element is that we integrate use-specific vertical costs and benefits into an open-city model in which wages, population, and land use are endogenously determined. Our aim is to demonstrate how the interplay of demand for space and use-specific net-costs of height shape skylines and horizontal land use patterns. Hence, we keep the household and firm location problem as simple as possible.[13] We provide a fuller discussion of the related theoretical literature, as well as potential for future research in Online Appendix Section C. Since we will use the model for quantitative evaluations throughout Sections 5 to 7, we develop the model using fairly restrictive, yet canonical functional forms.

Geography. We consider a linear city of endogenous length. One unit of land is available at each location, x, for development. The point $x = 0$ marks the historic center of the symmetric city, with $D(x) = |x|$ being the distance from the center.

Workers. Identical workers earn a city-specific wage, y. A worker living at floor s of a building located at x derives a Cobb-Douglas utility from the consumption of a tradable good, g, whose price is normalized to one and residential floor space f^R:

$$U(x,s) = A^R(x,s)\left(\frac{g}{\alpha^R}\right)^{\alpha^R}\left(\frac{f^R(x,s)}{1-\alpha^R}\right)^{1-\alpha^R}, \tag{1}$$

where $0 < (1-\alpha^R) < 1$ is the expenditure share on floor space. Utility depends on the location in vertical (s) and horizontal (x) space via $A^R(x,s) = \tilde{A}^R(x)s^{\tilde{\omega}^R}$, where $\tilde{A}^R(x) = \bar{a}^R e^{-\tau^R|x|}$. $\tilde{\omega}^R > 0$ is the height elasticity of residential amenity that captures height benefits, such as better views or less exposure to noise and pollution. $\tau^R > 0$ is a decay parameter that determines the rate at which utility declines in distance from the historic center and $\bar{a}^R$ moderates the utility at the center. Travel is inconvenient but free; hence, consumption is subject to the budget constraint $y = p^R(x,s)f^R(x,s) + g$, where $p^R(x,s)$ is the residential floor space rent.[14] One way to rationalize this setting is to assume that residents make a daily shopping trip by public transit to the historic center to purchase the tradable good and stop along the way for work. Utility maximization delivers the Marshallian demand functions $g = \alpha^R y^R$ and $f^R(x,s) = \frac{1-\alpha^R}{p^R(x,s)^{1-\alpha^R}}y^{\alpha^R}$ and the indirect utility $U(x,s) = A^R(x,s)\frac{y^R}{p^R(x,s)^{1-\alpha^R}}$. Assuming that residents are perfectly mobile within and across cities, utility is anchored to the reservation utility, which we normalize to $\bar{U} = 1$ without loss of generality. Setting $U(x,s) = \bar{U}$, we can solve for the residential bid rent, $p^R(x,s) = A^R(x,s)^{\frac{1}{1-\alpha^R}}(y^R)^{\frac{1}{1-\alpha^R}}$. Averaging across all floors of a building of height $S^R(x)$ at any location, x, we obtain the horizontal residential bid rent:

$$\bar{p}^R(x) = \frac{1}{S^R(x)}\int_0^{S^R} p^R(x,s)ds = \frac{a^R(x)}{1+\omega^R}S^R(x)^{\omega^R}, \tag{2}$$

where $a^R(x) = \tilde{A}^R(x)^{\frac{1}{1-\alpha^R}}(y^R)^{\frac{1}{1-\alpha^R}}$ and $\omega^R = \frac{\tilde{\omega}^R}{1-\alpha^R}$ is the height elasticity of residential rent. The average floor space demand at x is then given by $\bar{f}^R(x) = \frac{1-\alpha^R}{\bar{p}^R(x)}y^R$.

Firms. Atomistic firms produce the tradable good using labor, l, and commercial floor space, f^C, in a Cobb-Douglas production function:

$$g(x,s) = A^C(x,s)\left(\frac{l}{\alpha^C}\right)^{\alpha^C}\left(\frac{f^C(x,s)}{1-\alpha^C}\right)^{1-\alpha^C}, \tag{3}$$

where $0 < (1-\alpha^C) < 1$ is the factor share of floor space. Productivity is shifted in vertical (s) and horizontal (x) space by $A^C(x,s) = \tilde{A}^C(x)s^{\tilde{\omega}^C}$, where $\tilde{A}^C = \bar{a}^C N^{\beta} e^{-\tau^C|x|}$. $\tilde{\omega}^C > 0$ is the height elasticity of production amenity that captures height benefits, such as signaling and workplace amenity effects (Liu et al., 2018). The agglomeration elasticity of productivity $\beta > 0$ determines how productivity increases in city employment N (Combes and Gobillon, 2015). $\tau^R > 0$ is a decay parameter that determines the rate at which productivity declines in distance from the historic center and $\bar{a}^R$ moderates the productivity at the center. One way to rationalize this setting is to assume that all workers have to meet at the historic center to exchange knowledge. Maximization of the profit function $\pi(x,s) = g(x,s) - y^C l - p^C(x,s)f^C(x,s)$ delivers the marginal rate of substitution $\frac{l(x,s)}{f^C(x,s)} = \frac{\alpha^C}{1-\alpha^C}\frac{p^C(x,s)}{y^C}$, which is used in the profit function and assuming perfect competition and zero profits delivers the commercial bid rent $p^C(x,s) = A^C(x,s)^{\frac{1}{1-\alpha^C}}(y^C)^{\frac{\alpha^C}{\alpha^C-1}}$. Averaging across all floors of a building with height $S^C(x)$ at any location x, we obtain the horizontal commercial bid rent:

$$\bar{p}^C(x) = \frac{1}{S^C(x)}\int_0^{S^C} p^C(x,s)ds = \frac{a^C(x)}{1+\omega^C}S^C(x)^{\omega^C}, \tag{4}$$

where $a^C(x) = \tilde{A}^C(x)^{\frac{1}{1-\alpha^C}}(y^C)^{\frac{\alpha^C}{\alpha^C-1}}$ and $\omega^C = \frac{\tilde{\omega}^C}{1-\alpha^C}$ is the height elasticity of commercial rent.

Developers. Each unit of land at x can be occupied by one building that can house one use $U \in \{C, R\}$, where C indexes commercial and R indexes residential use. This assumption is consistent with tall buildings being highly specialized in terms of use (see Online Appendix Section C.1 for evidence). Perfectly malleable buildings of height, S^U, are constructed by developers facing the profit function:

$$\pi^U(x, S^U(x)) = \bar{p}^U(x)S^U(x) - \tilde{c}^U(S^U(x))S^U(x) - r^U(x), \tag{5}$$

where $\tilde{c}^U = c^U S^U(x)^{\theta^U}$. The baseline per-unit construction cost of a one-story building, c^U, is inflated in height by the height elasticity of construction cost, θ^U, which reflects that taller buildings require more sophisticated structural engineering and facilities, such as elevators. r^U is the land rent. Using Eqs. (2) and (4) in Eq. (5), we obtain the profit-maximizing building height $S^{*U}(x) = \left(\frac{a^U}{c^U(1+\theta^U)}\right)^{\frac{1}{\theta^U-\omega^U}}$. We require $\theta^U > \omega^U$ and $a^U > c^U(1+\theta^U)$ for a well-behaved solution to the height problem (at least one story is built). The chosen height by the developer is

$$\tilde{S}^U(x) = \min(S^{*U}(x), \bar{S}^U), \tag{6}$$

where $\bar{S}^U$ is a height limit that may be imposed by the planning system. Using $\tilde{S}^U(x)$ in Eq. (5), we obtain the use-specific bid rent for land under perfect competition and zero profits:

$$r^U(x) = \frac{a^U}{1+\omega^U}(\tilde{S}^U)^{1+\omega^U} - c^U(\tilde{S}^U)^{1+\theta^U} \tag{7}$$

Equilibrium. Land is allocated to the use associated with the highest land bid rent:

$$S^C(x) \quad = \tilde{S}^C(x), S^R(x) = 0, \text{ if } r^C(x) \geq r^R(x), r^C \geq r^A \tag{8}$$

$$S^R(x) \quad = \tilde{S}^R(x), S^C(x) = 0, \text{ if } r^R(x) > r^C(x), r^R \geq r^A, \tag{9}$$

where r^A is the agricultural rent. Under the restrictions $r^C(x=0) > r^R(x=0)$, $\frac{\partial r^U}{\partial D} < 0$, $\frac{\partial r^C}{\partial D} < \frac{\partial r^R}{\partial D}$, that are consistent with plausible parameter values, there are two points $\{-x_0, x_0\}$ at which the commercial and residential land rents equate ($r^C(\pm x_0) = r^R(\pm x_0)$). Between these points, commercial developers outbid residential developers when competing for land; thus these points define the boundaries of the central business district (CBD). Similarly, there are two points $\{-x_1, x_1\}$ where the residential and agricultural land rents intersect ($r^R(\pm x_1) = r^A$) and the city ends. Real estate markets clear, implying that all floor space supplied is input into either consumption or production:

$$F^C(x) \quad = S^C(x), x \in (-x_0, x_0) \tag{10}$$

[13] We refer the interested reader to the literature for a more realistic treatment of commuting (see e.g., Fujita and Ogawa, 1982; Lucas and Rossi-Hansberg, 2002; Ahlfeldt et al., 2015).

[14] This formulation is a hybrid between quantitative spatial models in which commuting costs are accounted for by a utility shift that exponentially depends on distance from the actual workplace location (Ahlfeldt et al., 2015) and canonical monocentric city models in which commuting costs increase linearly in distance from the CBD and affect utility indirectly via the budget constraint (Brueckner, 1987). Our choice is convenient because we obtain the same functional forms for the commercial and residential bid rents.

$$F^R(x) \quad = S^R(x), x \in (-x_1, -x_0) \cup (x_0, x_1), \tag{11}$$

where $F^C(x)$ is the total input of floor space of all firms at x and $F^R(x) = \bar{f}^R(x)n(x)$ is total floor space consumption by all workers $n(x)$ at x. Using $F^C(x)$ from Eq. 10, we obtain labor demand at each location via the marginal rate of substitution:

$$L(x) = \frac{\alpha^C}{1-\alpha^C}\frac{\bar{p}^C(x)}{y^C}S^C(x) \tag{12}$$

Using $F^R(x)$ from Eq. 11 in the Marshallian demand function, labor supply at any location is given by

$$n(x) = \frac{S^R(x)}{y^R}\frac{\bar{p}^R(x)}{1-\alpha^R}. \tag{13}$$

Finally, aggregate labor market clearing requires that

$$\int_{-x_0}^{x_0} L(x)dx = \int_{-x_1}^{x_0} n(x)dx + \int_{x_0}^{x_1} n(x)dx = N \tag{14}$$

We take parameters $\{\alpha^U, \beta, \omega^U, \theta^U, \tau^U, \bar{a}^U, \tilde{c}^U, \tilde{S}^U, \bar{U}, r^a\}$ as given and treat $\{y, N\}$ as city-wide endogenous objects for which we solve, along with the location-specific variables $\{L(x), n(x), \bar{p}^U(x), r^U(x), \tilde{S}^U(x)\}$, using the equilibrium conditions in Eq. (2), (4) and Eqs. (6) - (14), and a straightforward numerical procedure discussed in Online Appendix Section C.2. E

Intuition. The intuition behind our simple land use model is that greater amenity values of locations near the center translate into higher bid-rents and greater profit-maximizing building heights as long as the height elasticity of construction cost exceeds the height elasticity of rent ($\theta^U > \omega^U$).[15] Higher floor space rents and greater building heights lead to larger profits that fully capitalize into higher land rents. Since the general equilibrium of the model is defined by labor and land market clearing and free mobility anchors utility to a reservation level, an exogenous change in any parameter will generally lead to adjustments in all endogenous outcomes. We will explore the land use pattern the model generates in more detail in Section 5 and conduct several comparative static analyses in Sections 6 and 7, after we have chosen our preferred parameter values in Section 4.

4. Parameter values

To solve for the general equilibrium of the model introduced in Section 3, we need to settle on plausible values for its parameters $\{\alpha^U, \beta, \omega^U, \theta^U, \tau^U, \bar{a}^U, \tilde{c}^U, \tilde{S}^U, \bar{U}, r^a\}$. Because of the critical role the height elasticities $\{\omega^U, \theta^U\}$ play in shaping the equilibrium outcomes of the model and the related literature is in its infancy, we provide some estimates in Sections 4.1 and 4.2. Readers who are primarily interested in the application of the model may jump to Section 4.3 in which we discuss our preferred parameter values.

4.1. Costs of height

It is well established in the literature that the cost of supplying floor space increases in the density of development owing to diminishing returns to non-land inputs (Combes et al., 2021). In this section, we provide novel estimates of the use-specific elasticity of construction cost with respect to height θ^U and explore how innovations in construction technology may have affected the cost of height. For a review of the related literature, as well as a discussion of potential for future research, we refer to Online Appendix Section D.

[15] Regulation may result in a situation in which the profit-maximizing height, $S^*(x)$, is not realized and returns to height exceed the cost of height at the margin (Glaeser et al., 2005).

4.1.1. Estimating the cost of height

Using construction costs of a global cross-section of buildings provided by Emporis.com, we illustrate how the per-unit construction cost of commercial and residential buildings changes in height in Fig. 6, panel (a). Evidently, taller buildings are more expensive to build in per-floor space terms as the height elasticity of per-unit construction cost is positive.[16] Moreover, the elasticity increases in height. For buildings up to a height of nine floors, we estimate a height elasticity of 0.1. Beyond that point, the cost of height increases significantly.

Panel (a) in Fig. 6 not only suggests that the height elasticity of construction cost increases in height, but also suggests heterogeneity across uses. For taller buildings, we estimate a height elasticity of 0.26 for commercial buildings and about twice as large for residential buildings at 0.56. There are several reasons that may account for this difference. Residential units require more rooms with windows and, therefore, typically have smaller floor plates. Moreover, they need more sophisticated plumbing since residential units are equipped with bathrooms. Often, they have more complex facades with balconies (Smith et al., 2014).

There are few precedents in the literature to which we could compare our results. Most closely related to our estimates of the cost of height are Ahlfeldt and McMillen (2018) who also exploit the Emporis data set. Employing a more restrictive parametric estimation approach and extending the sample to very tall buildings, they estimate somewhat larger height elasticities of construction costs. Outside economics, there is a literature that provides engineering cost estimates. The rule of thumb is that construction costs tend to increase by 2% per floor (Environment, 1971), which is in line with more recent estimates (Tan, 1999; Lee et al., 2011).

4.1.2. Quantifying the effects of innovation on the cost of height

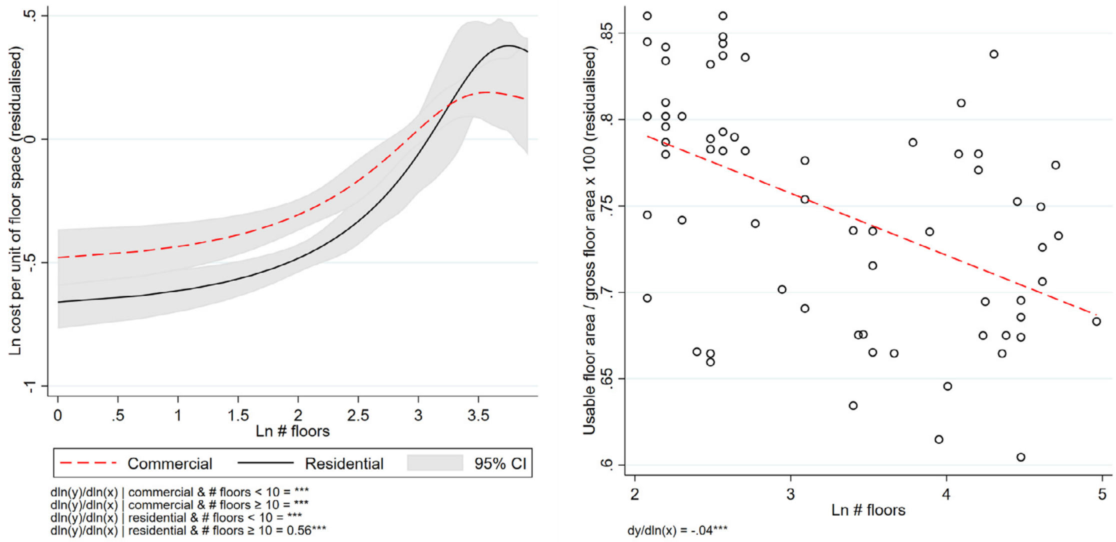
As discussed in some detail in Online Appendix Section B.1, the history of skyscrapers is marked by technological innovations. Quantifying the effect of technological innovations on the cost of height is empirically challenging since historical records of construction costs are not easily accessible. Within the structure of our model, the height elasticity of construction cost, θ^U, maps directly to the elasticity of height with respect to land rent $\kappa^U = \frac{\partial \ln S^U}{\partial \ln r^U} = \frac{1}{1+\theta^U}$ (see Online Appendix Section C.3 for a derivation). In a more general framework, Ahlfeldt and McMillen (2018) obtain $\kappa^U = \frac{\partial \ln S^U}{\partial \ln r^U} = \frac{\sigma^U}{1+\theta^U-\lambda^U}$. Our model features a special case in which the elasticity of substitution between land and capital is $\sigma^U = 1$ and the fraction of the parcel that is developable is independent of height (the height elasticity of extra space is $\lambda^U = 0$).

In Fig. 7, we provide estimates of κ^U for New York City and Chicago at different points in time from 1870 to 2010. Following Ahlfeldt and McMillen (2018), each estimate is derived from a building-level regression of the log of building height against the log of land value, using distance from important points of interest as instruments.[17] In line with our theoretical expectation, we find that the elasticity estimates increase over time. It is noteworthy that there were major reforms in the zoning regimes in Chicago in 1920, 1923, and 1957, which likely affected the relationship between building heights and land rents. Unlike for Chicago, the trend for New York is quite smooth across those dates.

We use our estimates of κ^U to recover decade-specific estimates of $\theta^U = \frac{\sigma^U-(1-\lambda^U)\kappa^U}{\kappa^U}$ under the more restrictive parametrization from our model ($\sigma^U = 1, \lambda^U = 0$), as well as borrowing parameter values $\sigma^C = 0.66, \sigma^R = 0.61, \lambda^C = 0.15, \lambda^R = 0.1$ from Ahlfeldt and McMillen (2018). We then regress the log of θ^U against a trend variable, weighting observations by the inverse of the standard errors in Fig. 7. Our tenta-

[16] We obtain the estimate of the height elasticity of construction cost from a regression of the log of construction cost per floor space unit (stripped of country and decade fixed effects) against the log of floor count.

[17] We refer to Fig. 7 notes for additional detail on the methodology and underlying data.



(a) Unit cost vs. height (b) Usable floor area vs. height

Fig. 6. Cost of height Note: In panel (a), we first regress the log of the ratio of building construction cost over building floor space against decade fixed effects, country fixed effects, and number-of-floors fixed effects. The displayed non-linear functions are the outcome of locally weighted regressions of the estimated number-of-floor fixed effects against the number of floors, using a Gaussian kernel and a bandwidth of 10. Confidence bands are at the 95% level. The vertical line marks the natural log of 10. Data are from https://www.emporis.com/ (see Ahlfeldt and McMillen (2018) for details). In panel (b), each marker is an individual building. The dependent variable is residualized in regressions against country fixed effects and a time trend. Note that the natural log of 10 (floors) is about 2.3. Observations were culled from Sev and Özgen (2009), Watts et al. (2007), Kim (2004), and Berger (1967).

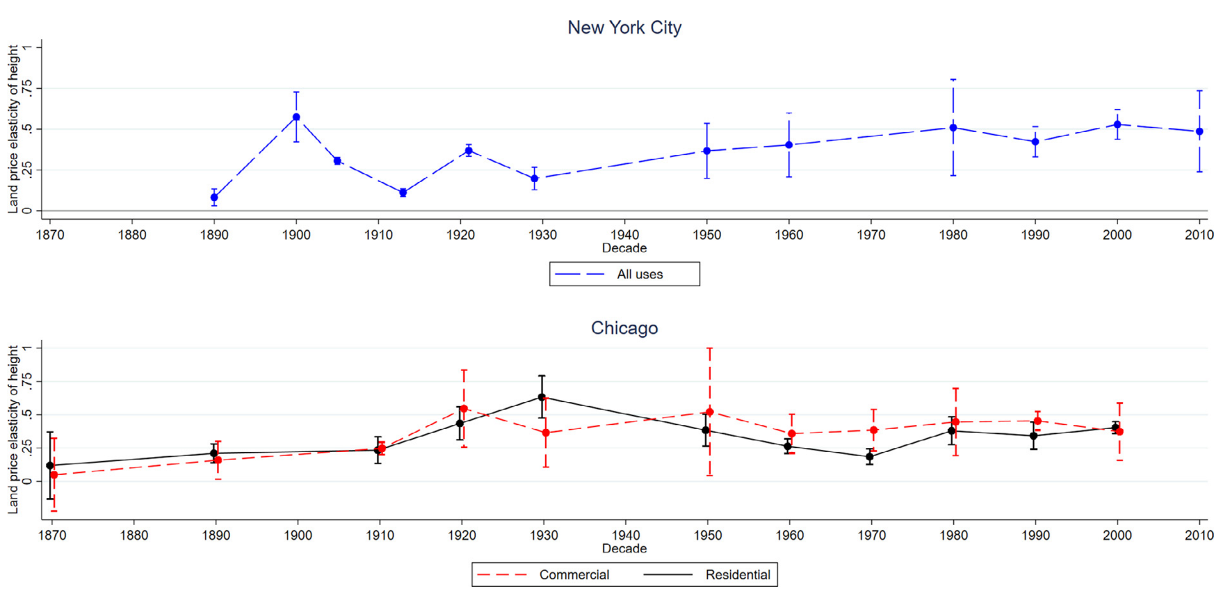


Fig. 7. Elasticity of height with respect to land price in New York and Chicago Note: Each point estimate is from a separate regression of the log of height of buildings constructed over a decade against the log of land value at the beginning of the decade, using the following instrumental variables. Log distance from Empire State Building and log distance from Wall Street for New York; Log distance from CBD and log distance from Lake Michigan for Chicago. 1940 missing in both data sets due to lack of completions. 1880 and 1900 missing for Chicago due to missing land value data. 1970 effect for New York is an imprecisely estimated outlier that was dropped to improve readability. Chicago data are from Ahlfeldt and McMillen (2018). New York city land values are from Barr (2016) and Spengler (1930). New York building height data are from the 2018 NYC PLUTO file from the NYC Dept. of City Planning.

tive interpretation of the results is that the cost of height decreased by about 2% per year over the course of the 20th century, although we stress that this interpretation hinges on assuming constant values for the elasticity of substitution between land and capital and the height elasticity of extra space. We refer to Online Appendix Section D.1 for further details.

4.2. *Returns to height*

Traditional urban economics models focus on how accessibility leads to variation in firm profits and household utility that capitalizes in horizontal rent gradients. However, productivity and amenities may also vary within buildings, leading to vertical rent and density gradients. In

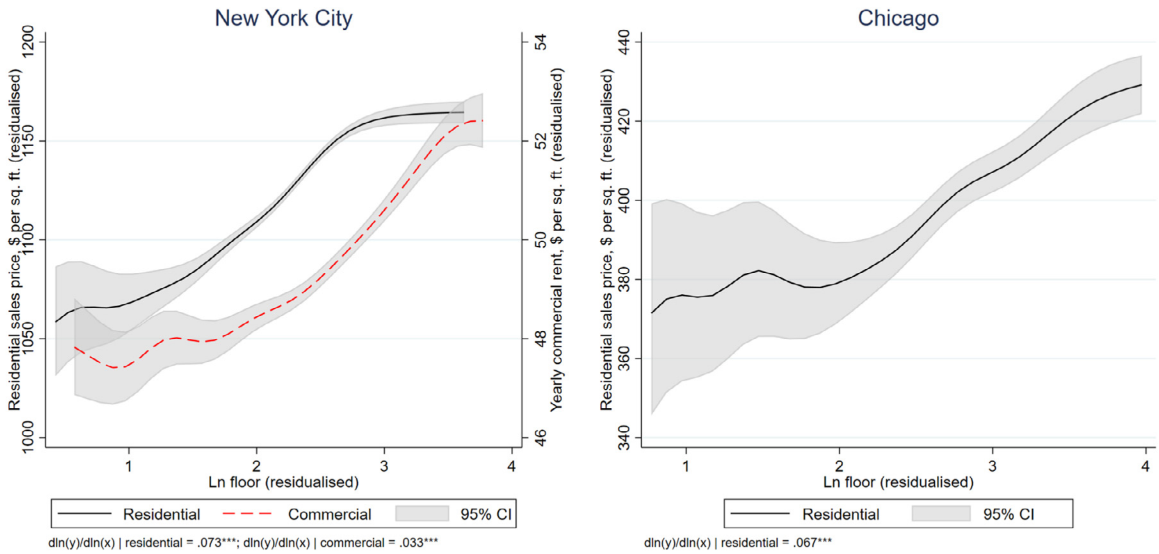


Fig. 8. Returns to height Note: Price, rent and floor are residualized in regressions (in logs) against unit characteristics and building fixed effects and time effects. The displayed non-linear functions are the outcome of locally weighted regressions using a Gaussian kernel and a bandwidth of 0.2. Confidence bands are at the 95% level. We trim the data set to exclude outliers that fall into the bottom or top percentile in terms of residualized ln rent/price or residualized ln floor before estimating the non-parametric gradient to improve the presentation. The parametric elasticity estimates are from the full sample. Residential prices are from StreetEasy. Commercial asking rents are from Cushman & Wakefield.

the model introduced in Section 3, developers face returns to height that are monitored by ω^U, the height elasticity of floor space rent. In this section, we provide estimates of ω^U and explore whether there is a time trend. For a review of the related literature, as well as a discussion of potential for future research, we refer to Online Appendix Section E.

4.2.1. *Estimating returns to height*

The commercial and residential bid rent functions derived in Section 3 establish a positive relationship between rent and height. It is straightforward to estimate ω^U by regressing the log of rent against the log of the floor at which a unit is located within a building, controlling for arbitrary building (and location) characteristics via building fixed effects.[18] Fig. 8 illustrates how the per-unit rent changes across floors within tall buildings in New York City and Chicago.[19] Evidently, there is a positive vertical height gradient. For commercial units, the positive height gradient can originate from increased revenues due to a signaling of quality to customers or a lower wage bill if height is a workplace amenity for which workers are willing to accept a compensating wage differential (Liu et al., 2018). For residential units, the rent increase likely reflects an amenity effect, e.g., due to better views. Note that we use purchase prices to mitigate effects of rent regulation, implicitly assuming that prices map to rents via a constant capitalization rate.

Reassuringly, we estimate an almost identical residential height elasticity of rent of about 0.07 for both cities.[20] For New York City, where we also have access to commercial rent data, we estimate a slightly lower commercial height elasticity of rent of 0.03. The somewhat lower height elasticity of rent for commercial than for residential buildings is consistent with luxurious condominiums occupying the upper floors in mixed-used skyscrapers, such as the Woolworth Building in New York or the Shard in London.[21]

Our estimates connect to a small literature that has estimated vertical height gradients within buildings, i.e., conditional on building fixed effects. For two samples of commercial buildings that cover various U.S. cities, Liu et al. (2018) estimate significantly larger values of the height elasticity of unit rent of 0.086 within the CompStat data set and even 0.189 within their Offering Memos data set. As for residential height premiums, Danton and Himbert (2018) estimate that within buildings in Switzerland, residential rents increase by 1.5% per floor. Their sample mostly consists of smaller structures. For the average floor of two within their sample, the implied height elasticity of unit rent is 0.03, somewhat less than what we find for taller residential structures in New York City and Chicago. One conclusion from relatively few existing estimates exploiting within-building variation is that the height elasticity of rent varies significantly across samples, land use, and, as we discuss next, time.

There are more estimates of the effect of the floor at which a unit is located on prices and rents from specifications that do not control for building fixed effects. While there is the concern that unobserved location attributes and within-building spillovers may confound the height effect, the results in this literature substantiate that the height elasticity of rent is positive (see Online Appendix Section E.2).[22]

It is worth recalling that in the context of our model, the height elasticity of rent ω^U is a reduced-form parameter that depends on the height elasticity of amenity $\tilde{\omega}^U = (1-\alpha^U)\omega^U$. Assuming the canonical values $1-\alpha^C = 0.15$ (Lucas and Rossi-Hansberg, 2002) and $1-\alpha^R = 0.33$ (Ahlfeldt and Pietrostefani, 2019), our estimates of ω^U imply that the

[18] See Online Appendix Section E.1.1 for a derivation of an estimation equation.

[19] We refer to Fig. 8 for a description of the underlying data.

[20] We obtain all elasticity estimates from regressions of log of rent or price against the log of the floor, controlling for unit characteristics, building fixed effects and time effects.

[21] The conclusion that returns to height are greater within residential than commercial buildings rests on the assumption that the capitalization rate of rents is independent of the floor.

[22] We note that for marginal changes, the height elasticity of unit rent discussed here and the height elasticity of average building rent derived in Eqs. (2) and (4) are the same. In reality, changes in height are typically non-marginal due to the integer floor constraint. In Online Appendix Section E.1.2, we show that for smaller buildings, suitable values of the height elasticity of average building rent that feed into the developer problem in Eq. (5) will typically be smaller than the values of the height elasticity of unit rent reported here.

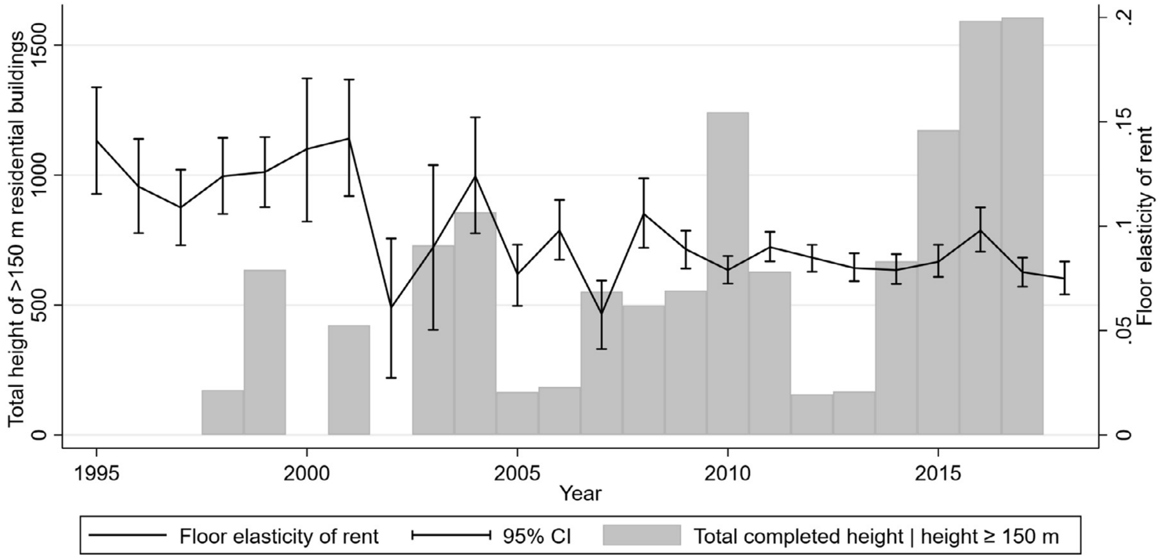


Fig. 9. Returns to height over time in the New York City residential market Note: The floor elasticity of rent is estimated in regressions of log rent against log floor by year, controlling for unit characteristics and building fixed effects. Confidence bands are at the 95% level. Residential rents are implied rents based on apartment prices data from StreetEasy, assuming a constant discount rate. Total completed height of residential buildings exceed 150 m are from the CTBUH Skyscraper Center.

productivity of firms, within a given building, increases in height at an elasticity of 0.005. Likewise, residential utility increases in height at an elasticity of 0.025.

4.2.2. Returns to height over time

Little is known about how the height elasticity of rent has changed over time. Fig. 9 summarizes year-specific estimates of the residential height elasticity of rent over 22 years in New York City. Subsequent to the September 11, attacks there was a sharp decline in the height premium in the subsequent two years. More generally, there is a general downward trend since the early 2000s that is seemingly at odds with the notion that skilled workers increasingly demand amenities (Glaeser et al., 2001). However, it is quite notable that the decrease in the height premium coincides with an expansion of the supply of tall residential buildings. The total height of completed residential buildings that are 150 m or taller sharply increased around the turn of the century. If there is heterogeneity in preferences, (implicit) prices are generally not independent of supply. One interpretation of the patterns observed in Fig. 9 is that there is significant dispersion in tastes for height. As the availability of the height amenity has increased, the valuation of the marginal buyer may have decreased. Alternatively, the amenity value of living on a high floor could have diminished since views are being obscured by additional tall buildings.

4.3. Baseline parameter values

Having reviewed the evidence on costs and returns to height in Sections 4.1 and 4.2, we are now ready to set the parameter values of the model developed in Section 3. We summarize our parameter choices in Table 1, where we also provide references to the related literature. These parameter values are not taken from individual papers and do not necessarily correspond to our own estimates. Instead, they represent what we view as canonical values that are suitable for stylized presentations and simple counterfactual analysis. We wish to highlight that the evidence base on the costs of and returns to height is small and parameter estimates of $\{\omega^U, \theta^U\}$ vary across samples and periods. Hence, some uncertainty surrounds our choices. That said, we provide various over-identification tests throughout the following sections that reinforce our belief that the chosen parametrization is sensible.

We use canonical values for the share of floor space as inputs $(1-\alpha^C)$, the share of floor space at consumption $(1-\alpha^R)$, and the agglomeration elasticity, β. We choose values of the height elasticity of construction cost that are in the ballpark of the estimates reported by Ahlfeldt and McMillen (2018), taking into account that the residential elasticity, θ^C, tends to be larger than the commercial elasticity, θ^R, (see Section 4.1). We acknowledge that the height elasticity of construction costs is likely increasing in height. Since we use the model for stylized predictions and illustrative counterfactual analyses, we stick to the parameterization with a constant elasticity for tractability. For the height elasticity of rent, we use our estimates reported in Section 4.2, taking into account that the residential elasticity, ω^R, appears to be larger than the commercial elasticity, ω^C. For the spatial decay in production amenity, τ^C, we choose a value based on novel estimates of the commercial rent gradient across a set of 55 global cities that complement recent evidence from the U.S. (Rosenthal et al., 2021) and are reported in Online Appendix Section F.1.1. This decay in production spillovers is conceptually different from the one estimated by Ahlfeldt et al. (2015), who assume that each location emanates spillovers to nearby areas. Guided by Ahlfeldt et al. (2015), we choose a smaller value for the residential amenity decay, τ^R.

5. Vertical and horizontal city structure

In this section, we solve our model to derive the vertical and horizontal structure of a stylized city. We contrast the predictions for the height gradient with novel evidence from a large set of cities around the world. For a review of the related literature, as well as a discussion of potential for future research, we refer to Online Appendix Section F.

5.1. Stylized gradients

Solving the model under the parameter values defined in Table 1 delivers the gradients illustrated in Fig. 10. The floor space rent for both uses naturally decrease in distance from the historic city center to compensate users for the associated loss of amenity. This is a standard prediction of spatial equilibrium models of internal city structure ever since Alonso (1964), if not von Thuenen (1826) We find that the commercial rent gradient is steeper than the residential rent gradient. An inspection of the first derivative of the log of floor space rent with respect to

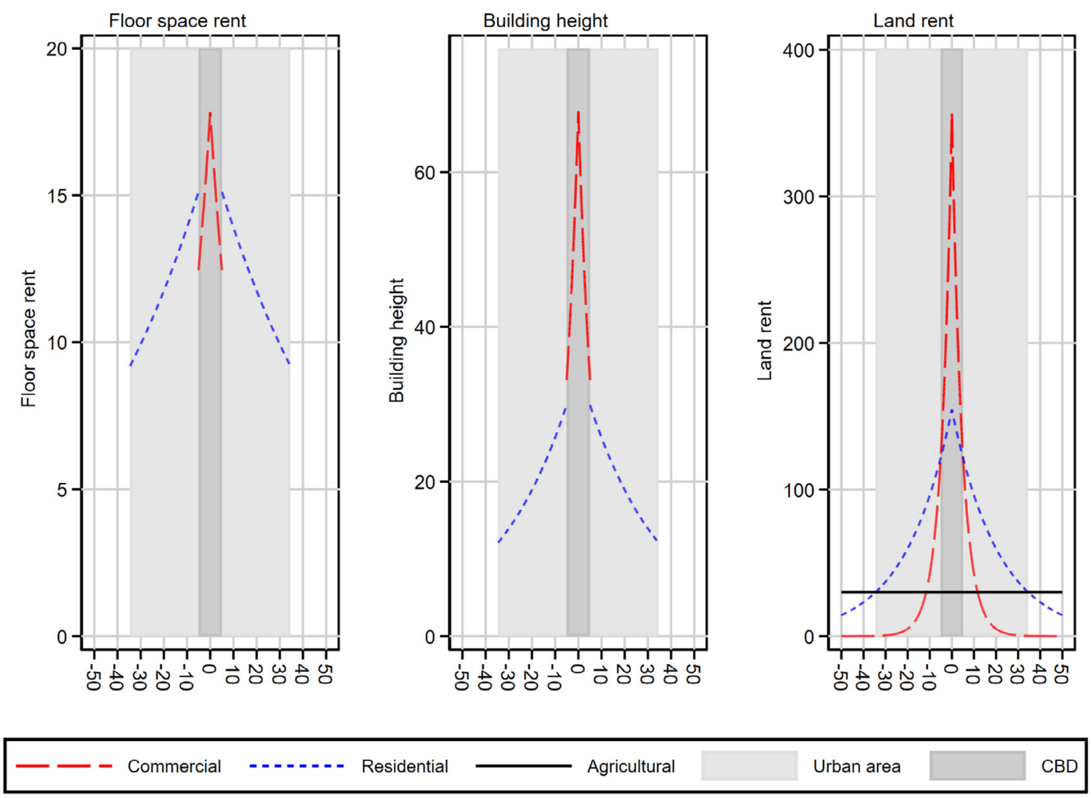


Fig. 10. Urban gradients Note: The figure illustrates the solution to the model laid out in Section 3 using the parameter values reported in Table 1. Floor space rent is the average per-unit rent within a building $\bar{p}^U(x)$. The x-axis gives the distance from the historic center in units that roughly correspond to kilometers in real-world cities.

Table 1
Parameter values.

	Parameter	Value	Further reading
$1-\alpha^C$	Share of floor space at inputs	0.15	Lucas and Rossi-Hansberg (2002)
$1-\alpha^R$	Share of floor space at consumption	0.33	Combes et al. (2019)
β	Agglomeration elasticity of production amenity	0.03	Combes and Gobillon, 2015
θ^C	Commercial height elasticity of construction cost	0.5	Ahlfeldt and McMillen (2018)
θ^R	Residential height elasticity of construction cost	0.55	Ahlfeldt and McMillen (2018)
ω^C	Commercial height elasticity of rent	0.03	Liu et al. (2018)
ω^R	Residential height elasticity of rent	0.07	Danton and Himbert (2018)
τ^C	Production amenity decay	0.01	Ahlfeldt et al. (2015)[a]
τ^R	Residential amenity decay	0.005	Ahlfeldt et al. (2015)

Notes: These parameter values are not taken from individual papers and do not necessarily correspond to our own estimates. Instead, they represent what we view as canonical values that are suitable for stylized presentations and simple counterfactual analysis. The last column provides a references for the interested reader for further reading, but not necessarily the source of a point estimate. [a]The parameter value is consistent with the commercial rent gradient estimated for a large set of global cities, assuming $\alpha^C = 0.15$ (see Online Appendix Section F.1.1). We set the following scale parameters arbitrarily to generate a plausible land use pattern: $\bar{a}^C = 2, \bar{a}^R = 1, \tilde{c}^C = 1.4, \tilde{c}^R = 1.4, r^a = 30, \bar{U} = 1$. There are no binding height limits in the baseline parametrization ($\bar{S}^C = \bar{S}^R = \infty$).

distance from the core $\frac{\partial \ln \bar{p}^U(x)}{\partial D(x)} = -\frac{\tau^U \theta^U}{(1-\alpha^U)(\theta^U-\omega^U)}$ reveals that the difference in the slope of the gradients is driven as much by the difference in amenity decay ($\tau^C > \tau^R$) as by the greater ease at which firms substitute away from floor space ($1-\alpha^C < 1-\alpha^R$). Due to the effect of the height amenity, the rent gradient also depends on the realized building height and, hence, the height elasticities, $\{\theta^U, \omega^U\}$, which is a novel theoretical insight. The chosen parameter values imply a semi-elasticity of building rent with respect to distance from the historic center of -0.071 for commercial use and -0.017 for residential use.

Higher floor space rents imply greater marginal returns to height and greater profit-maximizing building heights. Therefore, the height gradient follows the floor space gradient qualitatively. This is the standard prediction of neoclassical urban models incorporating a housing supply side (Mills, 1967; Muth, 1969), which has motivated Ahlfeldt and Barr (2020) to argue that tall historic buildings serve as proxies for historic centers that are often difficult to detect. The first derivative $\frac{\partial \ln S^* U}{\partial D} = -\frac{\tau^U}{(1-\alpha^U)(\theta^U-\omega^U)}$ reveals that the height gradient will be steeper than the floor space gradient unless the height elasticity of construction cost is set to implausibly large values, ($\theta^U \geq 1$). It is intuitive that the slope of the height gradient depends on the use-specific height elasticities, $\{\theta^U, \omega^U\}$. The chosen parameter values imply a semi-elasticity of height with respect to distance from the historic center of -0.142 for commercial use and -0.032 for residential use.

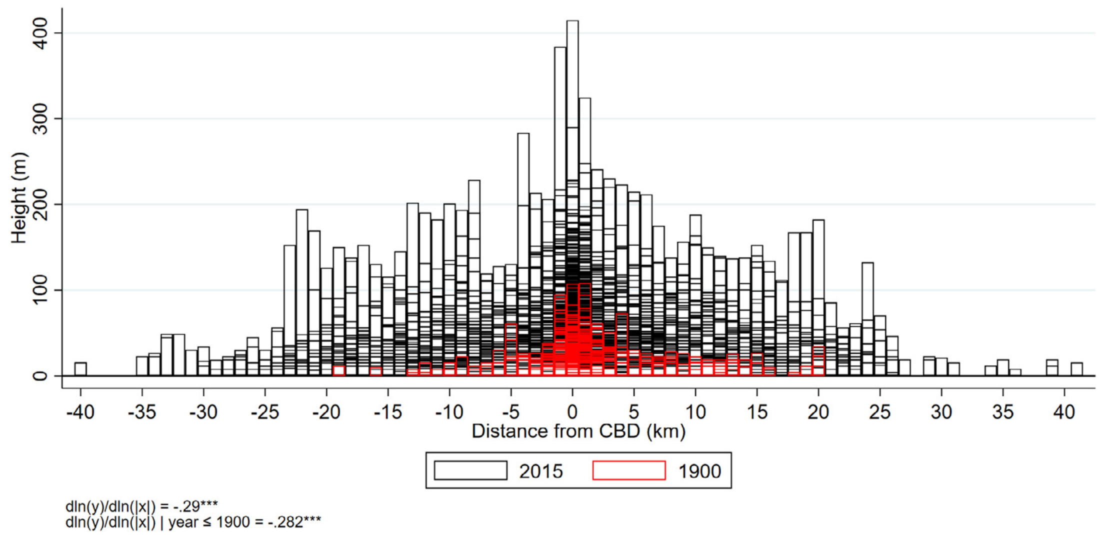


Fig. 11. Tallest buildings by North American city, and distance from the CBD Note: Each bar illustrates the height of the tallest building within a one-km bin to the west or the east of the CBD in one of 55 North American cities. Height data from Emporis. CBDs are the "global prime locations" identified by Ahlfeldt et al. (2020). Negative (positive) distance values indicate a location in the west (east) where the x-coordinate in the World Mercator projection is smaller (larger) than the x-coordinate of the CBD. Average height elasticities estimated conditional on city fixed effects. Data from https://www.emporis.com/ (see Ahlfeldt and McMillen (2018) for details).

The developer's ability to multiply land to floor space via height results in land bid rents that increase more than proportionately in floor space rents. Hence, Figure 10 illustrates how tall buildings rationalize extreme differentials in land prices across small areas, such as documented by Ahlfeldt and McMillen (2014) and Barr (2016). In fact, we generate an elasticity of land rent with respect to distance from the historic center of -0.53, which is within the range of estimates that Ahlfeldt and Wendland (2011) and Ahlfeldt and McMillen (2018) find for Berlin and Chicago (see Online Appendix Section F.1.2 for details).

Because land is allocated to the highest bidder, we can derive the horizontal land use pattern from the right panel of Fig. 10. The dark-shaded area where the commercial bid-rent for land exceeds the residential bid-rent marks the central business district (CBD). Similarly, the light-shaded area marks the residential zone where returns to residential development exceed the opportunity return in commercial and agricultural use. A novel prediction of our model is that, via their effect on floor space bid rents and profit-maximizing building heights, use-specific costs and returns to height affect the commercial and residential land rents and, hence, the horizontal land use pattern. This way, differences in the height elasticities, $\{\theta^U, \omega^U\}$, across uses have interesting feedback effects on the use-specific levels of floor space rents and building heights. An interesting feature of our parameterization is that the height and floor space gradients are discontinuous at the land use border. Since $\theta^R - \omega^R > \theta^C - \omega^C$, developers face greater net costs of height when developing residential rather than commercial buildings. Therefore, developers of commercial buildings just inside the CBD can afford to build taller at lower floor space rents than developers of residential buildings just outside the CBD. Indeed the discontinuities disappear if we set $\theta^R = \theta^C$ and $\omega^R = \omega^C$ (see Online Appendix Section F.1.3).

5.2. Evidence on height gradients

In Fig. 11, we pool the urban height profile of 55 North American cities sampled by Ahlfeldt et al. (2020). Consistent with the predictions of our model, building heights decrease relatively steeply in distance from the centre of the CBD. While, of course, absolute heights increased significantly, our estimates of the elasticity of height with respect to distance from the CBD, at -0.29, are almost identical for buildings that existed in 2015 and in 1900. In Table 2, we further zoom out to cover all 125 global cities for which Ahlfeldt et al. (2020) identify "global" prime locations (our CBD proxy). We focus on the tallest building within a one-kilometer distance ring from the CBD since we presumably measure the height of the tallest building with less error than the height of the average building. A comparison of columns 1 and 2, again, reveals a stability of the slope of the height gradient that is remarkable given the significant reductions in the cost of height suggested by the results in Section 4.1. One interpretation of the evidence is that, over the 20th century, reductions in the cost of height (monitored by θ^U) and the cost of CBD access (monitored by τ^U) have just about offset each other in their impact on the (relative) height gradient.

In columns (3) and (4) of Table 2, we find that the commercial height gradient is steeper than the residential height gradient. This finding is consistent with the model predictions under the chosen parametrization and can be rationalized by a relatively steep decay in the production amenity, a low input share of floor space in production, or a relatively low net cost of height in commercial development. A more specific prediction of the model is that the height gradient will be discontinuous if (and only if) costs and returns to height differ by land use. To test this prediction, we identify the edge of the CBD as the distance at which the residential building density starts exceeding the commercial building density in 15 large U.S. cities. Using a boundary discontinuity design, we estimate that building heights, on average, increase by 0.2 log points as one enters the CBD (see Online Appendix Section F.2 for details). This result provides indirect evidence for a relatively lower commercial height elasticity of construction cost, adding to the direct evidence in Section 4.1. The remaining columns of Table 2 illustrate how the height gradient is steeper in North American cities than in European or Asian cities. This suggests a role for height regulation, which we discuss in Section 7.

5.3. Fuzzy height gradient

Except for the height discontinuity at the land use boundary, all gradients in Fig. 10 are smooth. This is because we have summarized the amenity value of location by a smooth function of distance from the city center. This is a stark abstraction of vertical global cities where amenity values can vary remarkably within short distances due to scenic views, noise pollution, or access to subway stations. Since all (dis)amenities eventually capitalize into floor space rents, the profit-maximizing height can vary significantly over short distances, consistent with tall buildings standing side by side with smaller structures. An interesting question is whether our model, under the parameter values that we deem canonical,

Table 2
Height gradient estimates.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Ln(height)	Ln(height)	Ln(height)	Ln(height)	Ln(height)	Ln(height)	Ln(height)
Distance from	-0.069***	-0.042***	-0.052***	-0.036***	-0.024***	-0.023***	-0.048***
CBD center (km)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$d\ln(y)/d\ln(x)$	-.23	-.27	-.33	-.26	-.12	-.12	-.35
Building Sample	$t^a \leq 1900$	All	Comm.[b]	Residential	Asia	Europe	North A.[b]
Observations	344	3469	1294	1185	2081	662	419
R^2	0.494	0.559	0.679	0.647	0.427	0.331	0.376

Notes: [a]: Completion year. [b] Commercial. [c]: North America. Unit of observation is city-distance bin (1 km). Height is the height of the tallest building within a one-km distance bin. Building data from Emporis. CBD definitions (global prime locations) for 125 global cities from Ahlfeldt et al. (2020). Elasticities computed at the sample means. Data from Empiris (see Ahlfeldt et al. (2020) for details). * p < 0.1, ** p < 0.05, *** p < 0.01.

can rationalize real-world skylines with plausible variation in amenity values.

To answer this question, we follow the conventions in the literature on quantitative spatial models of internal city structure (Ahlfeldt et al., 2015) and i) discretize space, ii) introduce a location-use-specific structural fundamental that scales the amenity parameters, $\{\bar{a}^R, \bar{a}^C\}$, and iii) exploit the direct mapping from amenity to height established in Eq. (6) to invert the structural fundamental from building heights observed in data. Intuitively, we solve for the unobserved values of the fundamental amenity that make the observed values of building heights consistent with an equilibrium in our model.

We take Chicago as a case in point; most tall buildings are approximately linearly aligned along the shore of Lake Michigan. To create a stylized version of the Chicago skyline, we first generate a linear grid of 100-meter cells defined in terms of the northings (the y-coordinate in a projected grid reference system). The equivalent of x in our model is then simply the difference of the northing of a grid cell to the northing of the global prime location in Chicago, as identified by Ahlfeldt et al. (2020). We assign the height of the tallest building and its use observed in the Emporis data to the entire grid cell. The resulting stylized skyline, which we exactly match in our model, is in the upper panel of Fig. 12. Evidently, the model now accommodates a *fuzzy height gradient* with substantial micro-geographic variation in building heights. We also observe mixing of uses because of tall residential buildings located within the CBD where there are direct views on amenities, such as Lake Michigan or the Chicago River. Towards the edges of the simulated city, there are empty parcels, which we rationalize with a zero fundamental amenity in a theory-consistent way.

In the bottom-left panel, we illustrate the distribution of the fundamental amenity values that convert the smooth height gradients in Fig. 10 into the real-world fuzzy height gradient in the top panel of Fig. 12. The residential fundamental amenity scales utility by factors that range between about 1 and 1.3, with the great majority of locations falling into a ± 0.1 window centered on 1.1. The commercial fundamental amenity scales productivity by factors that fall into an even narrower range of about 0.85 to 1. Hence, relatively little variation in fundamental amenity is required to introduce sizable fuzziness into the height gradient. Intuitively, users substitute away from floor space as its price increases, leading to a more-than-proportionate increase in floor space rents as the amenity value increases. As the floor space rent increases, developers use relatively more capital to construct buildings whose height increases more than proportionately in floor space as long as $\{\theta^U - \omega^U\} < 1$. Therefore, substitution on the consumption side (by users) and production side (by developers) amplify small differences in amenity into large differences in height and land rent.

The bottom-right panel suggests that our model does a good job in describing these substitution patterns. Given the fundamental amenity values inverted from building heights, the model generates land rents that scale proportionately to land values observed in data (from Ahlfeldt and McMillen, 2018). The fit along the 45-degree line suggests that the model not only correctly predicts land rents to be higher where they tend to be higher in the real world, but also correctly predicts the relative magnitudes. This lends support to the chosen parametrization and substantiates the point made by Ahlfeldt and Barr (2020) who argue that tall historic buildings can be used to predict centers of economic activity in history.

The main conclusion from Fig. 12 is that relatively small differences, with respect to the micro-geographic amenity, rationalize large differences in heights of adjacent buildings, especially if there is mixed land use. Of course, there are additional explanations which we discuss in more detail in Online Appendix Section F.4.[23] Yet, a simple model can account for the shape of real-world skylines largely through differences in commercial and residential amenities, an important lesson for future quantitative spatial models that tackle the vertical dimension of cities.

6. Skyscrapers as causes and effects of agglomeration

Empirically, skyscrapers appear to be a distinctively urban phenomenon (see Section 2). Therefore, it is surprising that little research has been done to establish the relationship between urbanization and vertical growth in a systematic and robust manner. In this section, we explore theoretically and empirically how cities expand into the vertical dimension as the level of agglomeration increases and reductions in the cost of height foster agglomeration. For a review of the related literature, as well as a discussion of potential for future research, we refer to Online Appendix Section G.

6.1. Returns to agglomeration

One reason why cities exist is that agglomeration leads to greater productivity (Combes and Gobillon, 2015). Returns to agglomeration have likely risen over the 20th century (Moretti, 2012); this should have led to increasing demand for central city locations and the development of skyscrapers.[24] We subject this intuition to a quantitative evaluation

[23] Fragmentation of land ownership can prevent the assembly of sufficiently large lots to develop a skyscraper (Strange, 1995; Brooks and Lutz, 2016). The durability of tall structure can add to the fuzziness of skylines that consist of skyscrapers developed at different points in time (Brueckner, 2000), although we do not find the fuzziness of the height gradient is much reduced within construction date cohorts (see Online Appendix F.3). Similarly, delayed skyscraper development generated by option values (Titman, 1985; Williams, 1991) can lead to fuzziness. Finally, there is a battery of institutional factors, such as land use regulation, historic preservation, and public ownership, that can add to micro-geographic variation in building heights.

[24] Autor (2019) documents that over the recent decades this trend applies to high-skilled workers, exclusively. In contrast, the urban wage premium in the U.S. has sharply declined for low-skilled workers since the 1970s.

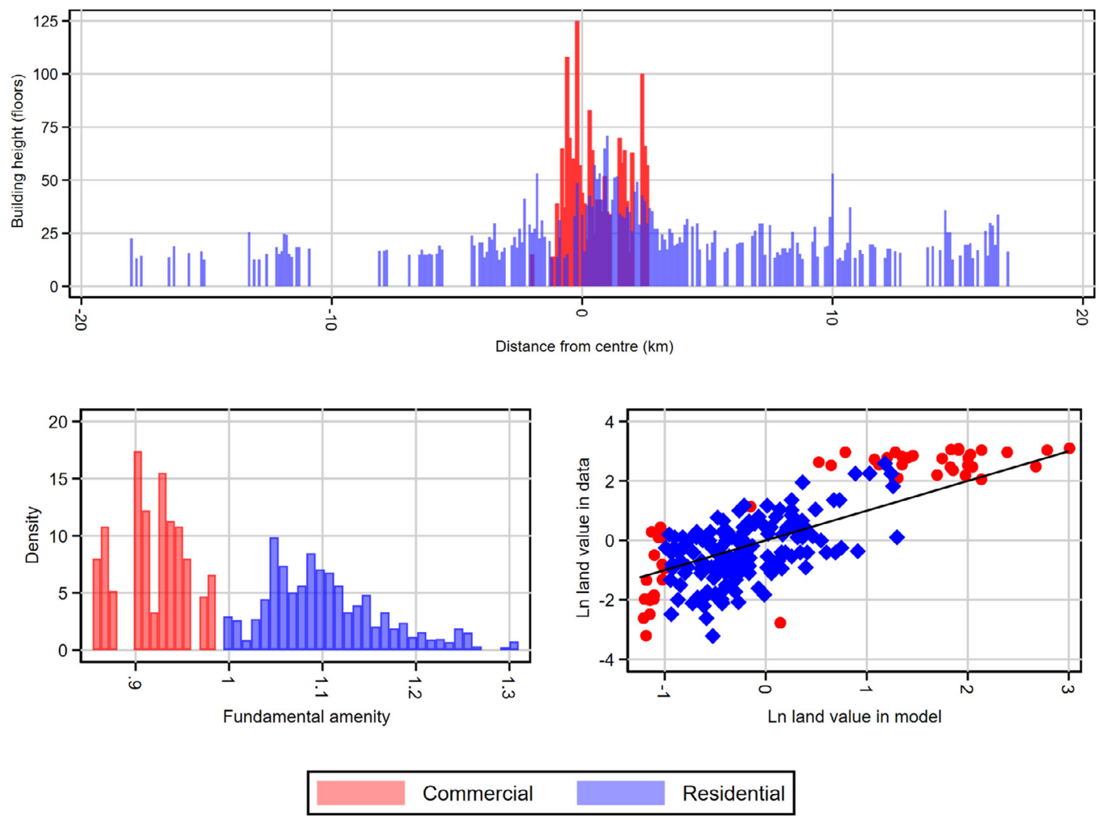


Fig. 12. Fuzzy height gradient Note: Figure illustrates the solution to the model laid out in Section 3 using the parameter values reported in Table 1), discretized space (to 100-meter cells) and a multiplicative structural amenity components that we invert such that the model matches the stylized skyline of Chicago in the top panel. Distance from the center is the difference between the northings of a grid cell and the northing of the global prime location identified by Ahlfeldt and McMillen (2018). The bottom-left panel shows the distribution of the inverted fundamental amenity. The bottom-right panel compares the model's predictions of land rent to 2000 land values from Ahlfeldt and McMillen (2018).

by solving the model developed in Section 3 for varying values of the agglomeration elasticity, β.

We correlate some of the endogenous model outcomes with the set values of β in Fig. 13. The left panels show how greater returns to agglomeration translate into higher wages, which leads to the attraction of a greater workforce. Because we follow the canonical spatial equilibrium framework and allow for free migration of homogeneous workers, the equilibrium population is highly sensitive to the choice of β. In response to increasing demand, the city expands horizontally (upper-middle panel) and vertically (right panels). Whereas the residential zone expands significantly into the former agricultural hinterland, the CBD grows primarily into the vertical dimension. Greater competition for space leads to higher equilibrium floor space rents since taller buildings are more expensive to build (bottom-middle panel). The city-size elasticity of floor space rent inferred from a comparison of equilibrium outcomes across simulation runs is about 0.3. This is what would be expected for a large and dense city according to the empirically derived rule-of-thumb by Ahlfeldt and Pietrostefani (2019), lending further support to our parametrization of the model.[25] This counterfactual exercise also establishes an elasticity to which we return with empirical estimates in Section 6.4: The city size elasticity of the tallest building height ranges from 0.52 (residential) to 0.66 (commercial). We refer to Online Appendix Section G.1 for further detail on those estimates.

6.2. Transport innovations

New transport technologies that started with horse-driven street cars, then improved to subways, and then automobiles and highways have significantly reduced within-city transport costs. It is well established in the literature that improvements in transport technology cause cities to grow and economic activity within cities to decentralize (Baum-Snow, 2007; Redding and Turner, 2015). Less is known about the implications on building heights. In our model, within-city transport cost is moderated by the amenity decay parameters, $\{\tau^C, \tau^R\}$. As already discussed in Sections 3 and 5, smaller values of $\{\tau^C, \tau^R\}$ lead to a flatter height gradient. The implications for absolute building heights are less straightforward since the relative increase in attractiveness of peripheral locations and the increase of the attractiveness of the city as a whole have countervailing effects on building heights in central locations. To illustrate how these conflicting forces play out in our model, we solve the model for varying values of the amenity decays, $\{\tau^C, \tau^R\}$. In each run, we set a different value for the commercial amenity decay τ^C and adjust the residential amenity decay such that the difference between the commercial and residential decay parameters conforms to the baseline, i.e., $\tau^R = \tau^C - 0.005$. This way, we only change one parameter value at a time, which allows us to present the results in Fig. 14 using the same structure as in Fig. 13.

As we increase transport costs, the city becomes less attractive. Hence, the population shrinks and wages fall, owing to lower agglomeration economies (left panels). In line with intuition, the city contracts in horizontal space (upper-middle panel) as transport costs increase. While the partial equilibrium effect would be a shift in demand towards the city center, resulting in an increase in floor space rents and building

[25] This value is also in line with estimates for large French cities (Combes et al., 2019).

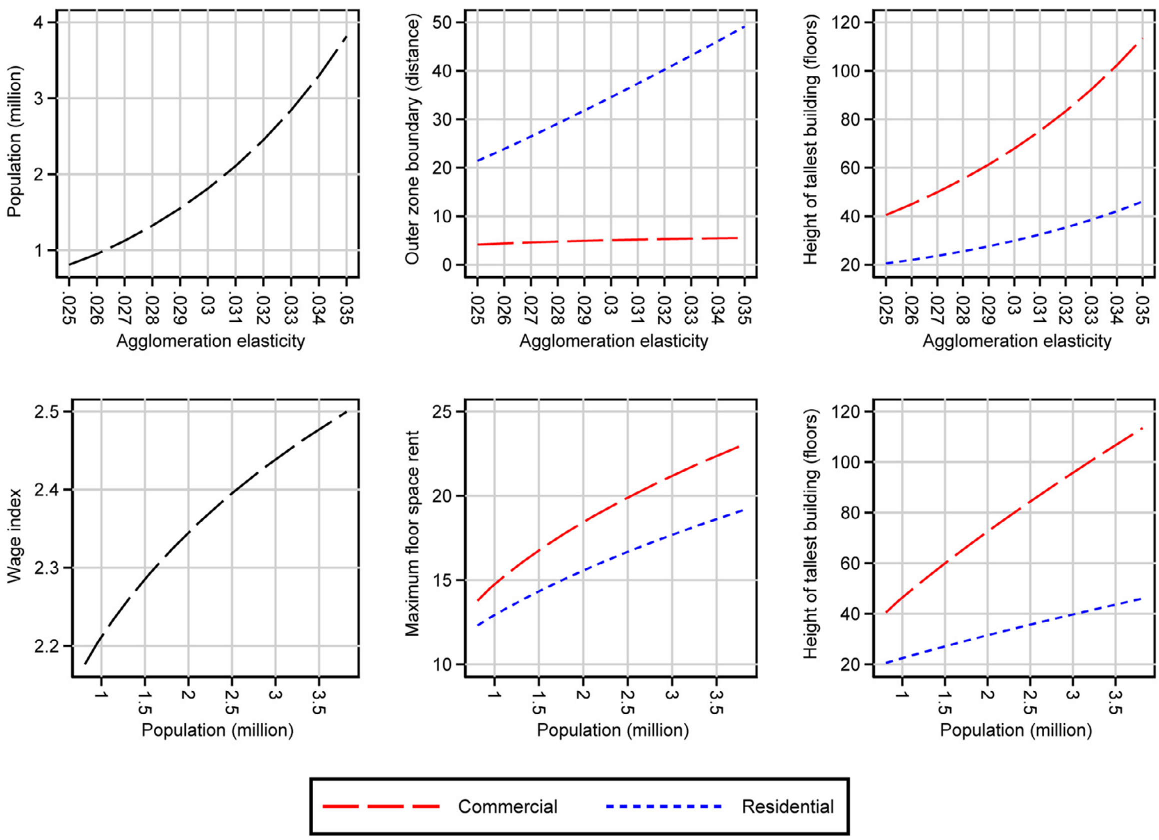


Fig. 13. Variation in external returns to agglomeration Note: We report solutions to the model developed in Section 3 under varying values of the agglomeration elasticity, β. All other parameter values are kept constant at the levels reported in Table 1.

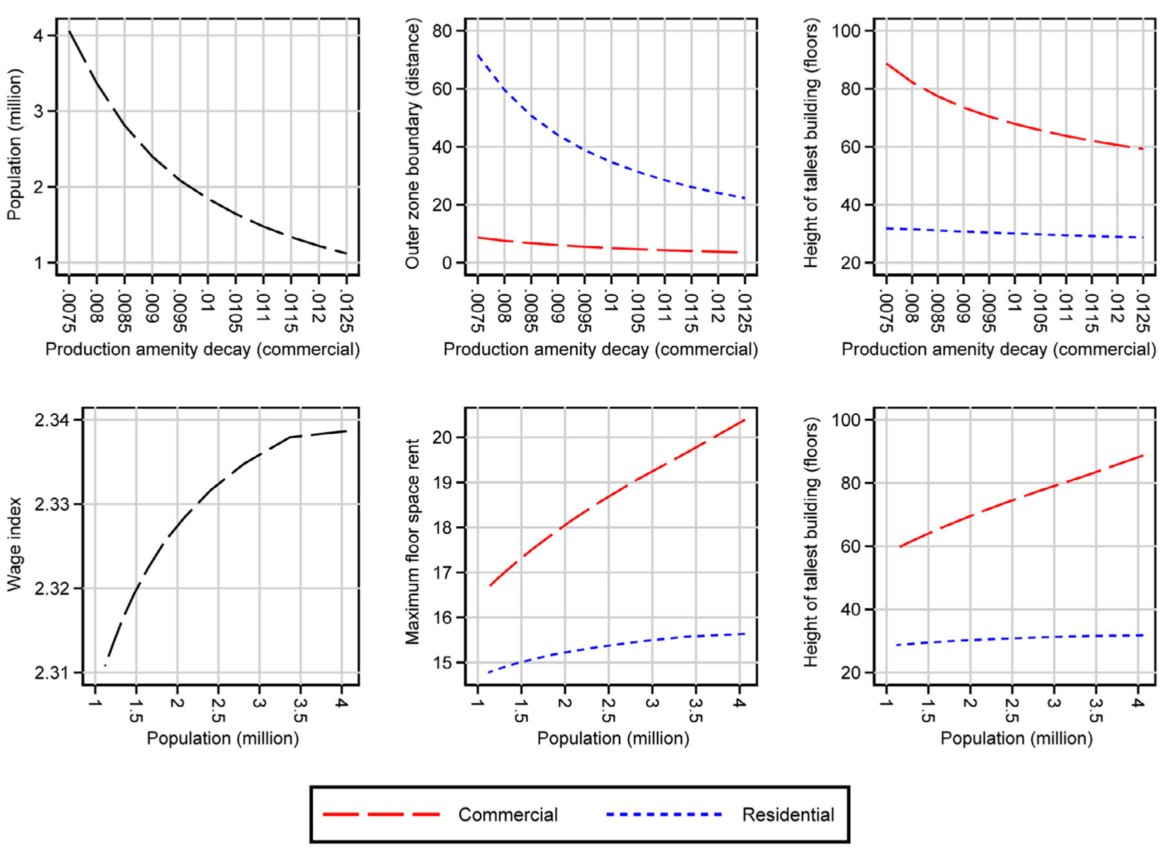


Fig. 14. Variation in transport technology Note: We report solutions to the model developed in Section 3 under varying values of the amenity decay parameters $\{\tau^C, \tau^R\}$. We keep the relative size constant at $\tau^C - \tau^R = 0.005$. All other parameter values are kept constant at the levels reported in Table 1.

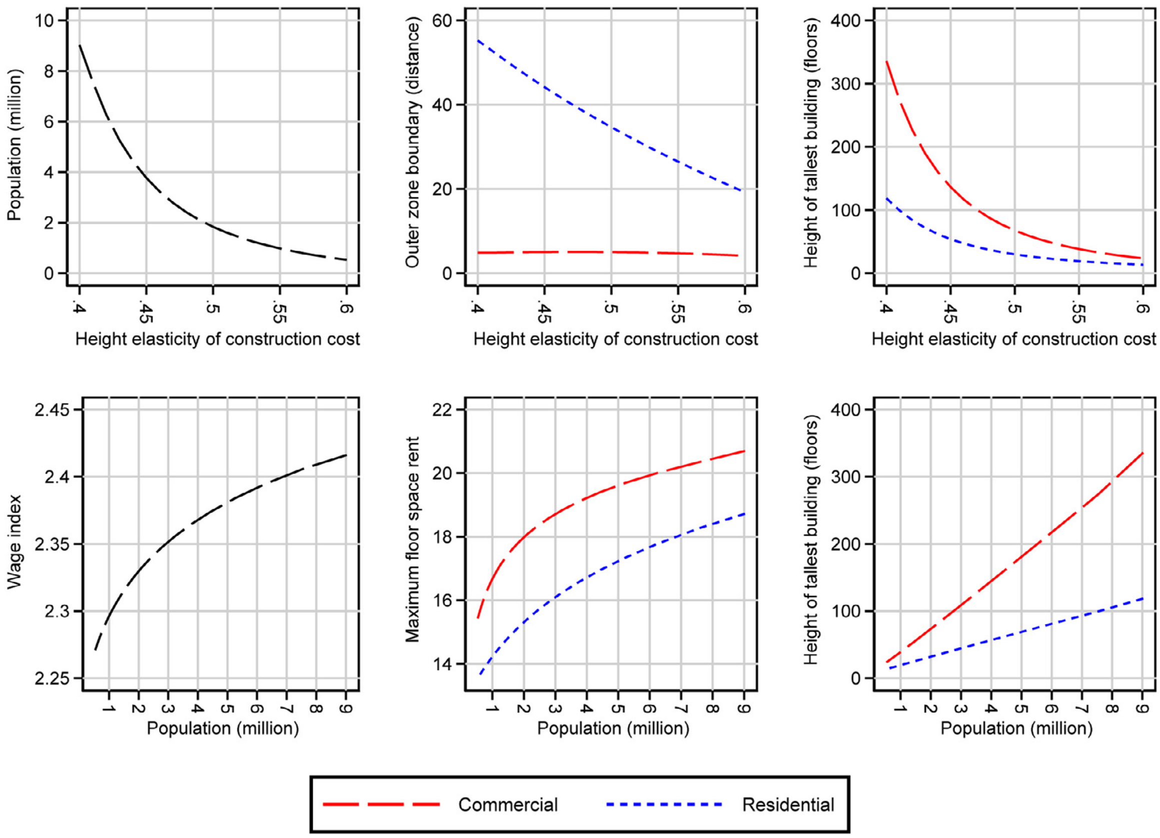


Fig. 15. Variation in cost of height Note: We report solutions to the model developed in Section 3 under varying values of the amenity decay parameters $\{\theta^C, \theta^R\}$. We keep the relative size constant at $\theta^C - \theta^R = 0.05$. All other parameter values are kept constant at the levels reported in Table 1.

heights in central locations, the general equilibrium effect is the opposite. The city-wide decrease in population and wage more than offsets for the relative increase in demand for central locations, so that commercial building heights decrease (upper-right panel). The effect on residential heights is qualitatively similar, but quantitatively smaller since the adjustment in the horizontal size of the zone is much greater. Because the increase in population as we reduce transport cost is driven by a horizontal expansion, we find relatively low city size elasticities of rent and building height. The city size elasticity of rent ranges from 0.05 (residential) to 0.15 (commercial). The city size elasticity of tallest building height ranges from 0.08 (residential) to 0.3 (commercial). We refer to Online Appendix Section G.1 for further detail on those estimates.

6.3. *Construction technology*

Various technological innovations discussed in Section 2 and Online Appendix Section B.1 have led to a reduction in the cost of constructing tall buildings since the end of the 19$^{\text{th}}$ century. The results in Section 4.1 point to a significant reduction in the height elasticity of construction cost. To evaluate the effects of this important, yet understudied, technological trend on urban structure, we solve our model for varying values of the height elasticity of construction cost $\{\theta^C, \theta^R\}$. As with the amenity decay in Section 6.2, we select a different value for the commercial height elasticity of construction cost, θ^C, in each run and adjust the residential residential elasticity such that the difference between the commercial and residential elasticity conforms to the baseline, i.e., $\theta^R = \theta^C - 0.005$. This way, we only change one parameter value at a time, which allows us to present the results in Fig. 15 using the same structure as in Figs. 13 and 14.

A reduction in the cost of height implies that developers will supply more floor space for a given rent level. Ceteris paribus, this would reduce equilibrium rents and increase the attractiveness of the city. In our open-city model, population adjusts to restore the exogenous reservation utility. As the population grows, productivity, and wages increase (left panels). To accommodate the larger population, the residential zone expands in horizontal and vertical space. The CBD expands vertically, primarily (upper-middle and upper-right panels). Since profits and utility are anchored to exogenous levels, rents adjust to offset for the increased productivity and wage (bottom-middle panel). Under the chosen parameterization, developers respond to the increased population, productivity and wages by building tall buildings. The city size elasticities of building height, at 0.77 (residential) and 0.94 (commercial), are large. As a result of the rapid expansion of supply of floor space, the city size elasticities of rent, at 0.1 (commercial) and 0.12 (residential), are relatively low (see Online Appendix Section G.1 for further detail on those estimates). These magnitudes, however, need to be taken with a grain of salt. Our parameterization with a constant height elasticity of construction cost is arguably a reasonable approximation for moderately tall buildings. The elasticity is significantly larger at extreme heights (Ahlfeldt and McMillen, 2018). Without affecting the qualitative implications, we would likely observe a more concave relationship between population and tallest building height if we parametrized the height elasticity of construction cost as a positive function of building height.

6.4. *Empirical implications*

One unambiguous implication that emerges from Figs. 13, 14, and 15 is that we should expect a positive relationship between population and the vertical size of cities, theoretically. We subject this model prediction to an empirical test in Fig. 16. Indeed, we estimate a positive city size elasticity of tallest building height of 0.25 for a set of international cities with population greater than one million. For smaller cities, there is no significant correlation in the data. We find almost the

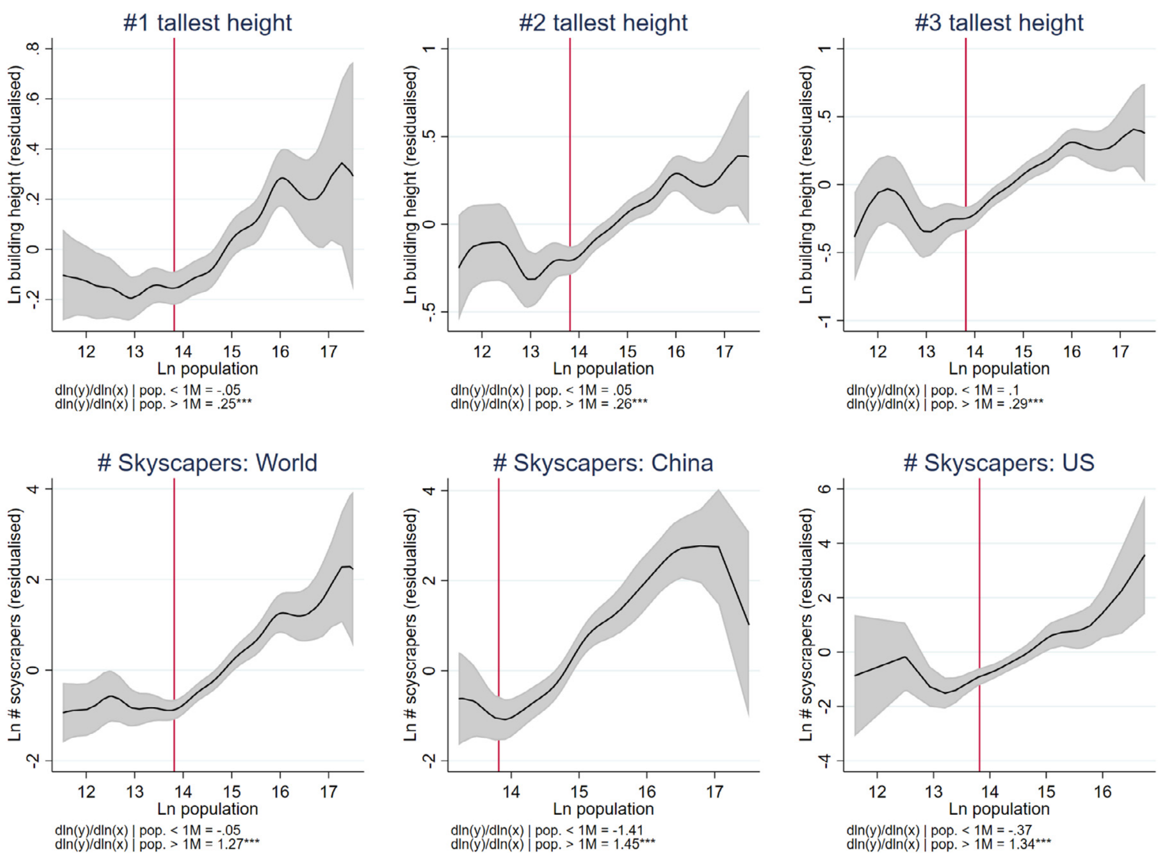


Fig. 16. Skyscrapers and city size Note: Skyscrapers and population for cities around with world, with at least one 150 m+ skyscraper. ln(building height) and ln(# skyscrapers) are residualized in regressions against country fixed effects. The black solid lines are from locally weighted regressions using a Gaussian kernel and a bandwidth of 0.25. Confidence bands are at the 95% level. Vertical line marks the log of one million. Skyscraper data is from https://www.skyscrapercenter.com/; accessed Feb. 2020. Population data are from several sources (available upon request) and are the most currently available counts for the metropolitan regions, which includes the central city and surrounding population agglomerations.

identical pattern if we consider the second or third tallest building in the city, which suggests that the relationship is not driven by outliers. For large cities, there is also a robust relationship between city size and the number of skyscrapers (150 m+). The implied city size elasticity of the per capita number of skyscrapers is $1.27 - 1 = 0.27$. We estimate a somewhat larger value of 0.34 when restricting the sample to cities in the U.S. This is slightly less than the elasticity of 0.38 estimated by Albouy et al. (2020) for U.S. cities. For China, the estimate is even larger. Again, for cities with a population of less than one million, there is no significant correlation. It appears that there is a size-threshold beyond which the skyscraper technology becomes viable or legally allowed.

In interpreting the reduced-form relationship between city size and building heights, it is important to acknowledge that we have generated this relationship via three distinct channels in Figs. 13, 14, and 15, each of which relates to one major long-run trend. The implication is that, empirically, the magnitude of the city size elasticity of building height will depend on what is driving variation in city size in the data. Given the parameterization of our model, if city size variation comes from transport improvements, we will estimate a lower city size elasticity with respect to building heights as compared to variation coming from the cost of height. Tentatively, we can conclude that the magnitudes estimated from the data suggests that the relationship between city size and building heights observed in our data is likely driven by a mix of both channels.

There are also important implications for the direction of causality. In Figs. 13 and 14, we manipulate the demand side of land markets and this causes the supply of tall buildings to respond. Intuitively, urbanization causes skyscraper development. To establish this causal relationship empirically, one would require variation in population that is unrelated to the supply side of land markets. The same logic motivates the use of Bartik (1991) instruments in the estimation the housing supply elasticity (Saiz, 2010). In Fig. 15, we manipulate the supply side of land markets and this causes the demand side to respond. Intuitively, skyscraper development causes urbanization. To establish this causal relationship empirically, one would require variation in population that is unrelated to the demand side of land markets. This logic underpins the use of geological conditions as instruments for density in the identification of agglomeration spillovers (Rosenthal and Strange, 2008; Combes et al., 2011), although the role of bedrock as a facilitator of skyscraper development has been challenged (Barr et al., 2011; Barr, 2016).

7. Height regulation

So far, we have treated land markets as competitive and unregulated. Of course, cities impose various forms of zoning and building regulations that affect height decisions (Brueckner and Sridhar, 2012). In this section, we first use our model from Section 3 to derive how height limits change the vertical and horizontal structure of a city (Section 7.1). We then turn to the welfare implications and how our open-city approach complements recent research that—implicitly or explicitly—refers to the closed-city model (Section 7.2). For a review of the related literature as well as a discussion of potential for future research, we refer to Online Appendix Section H.

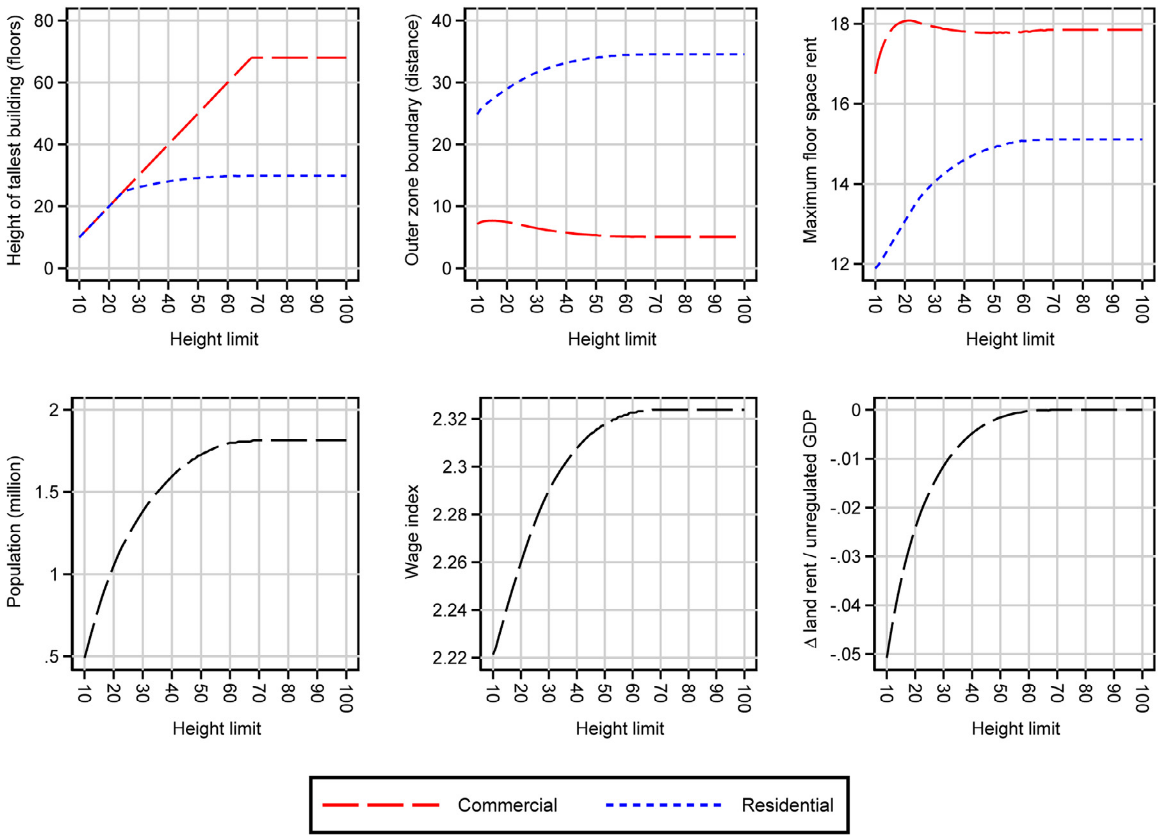


Fig. 17. Variation in height limit Note: We report solutions to the model developed in Section 3 under varying values of the height limit, $\bar{S}^R = \bar{S}^C$. All other parameter values are kept constant at the levels reported in Table 1. In the bottom-left corner, we measure aggregate land rent over the area occupied by the unregulated city. We do not vary this interval across simulation runs and aggregate over the maximum land bid rent, $\max(r^C(x), r^R(x), r^a(x))$, at each location. GDP is the wage bill, (yN), weighted by the inverse of the labor factor share, α^C.

7.1. Effects on horizontal and vertical structure

Our model developed in Section 3 allows for use-specific height limits, $\bar{S}^U$, which can be set by a planner. As per Eq. (6), these height limits will be binding if $\bar{S}^U < S^{*U}$, where S^{*U} is the profit maximizing building height that we have evaluated thus far. To evaluate the effects of a height limit on urban structure, we solve our model for varying values of a height limit, $\bar{S} = \bar{S}^C = \bar{S}^R$, which we apply uniformly to commercial and residential buildings for simplicity. We correlate a range of endogenous model outcomes with the set values of $\bar{S}$ and in Fig. 17.

There is ample evidence that real-world height limits tend to be binding (see Online Appendix Section H.2). We know from Fig. 10 that the tallest building in our unregulated stylized city is slightly smaller than 70 floors. Hence, consistent with reality, only very generous height caps do not affect the spatial structure of the city. An interesting insight from Fig. 10 is that, although the tallest residential building in the unregulated city has 30 floors, even height limits greater than 30 floors have an indirect effect on residential building height. This is because unless the height limit reaches the greatest profit-maximizing commercial building height, it constrains the commercial sector, and reduces labor demand, wages, population, and, eventually, demand for residential floor space.

Height limits also have an effect on the horizontal land use pattern. A height limit reduces the amount of commercial floor space at every location where it binds. The limit to vertical expansion results in the CBD growing horizontally, pushing the land use boundary and residential use outwards. The effect on the horizontal size of the residential zone is the opposite. This is because the reduction in the productivity of the commercial sector due to displacement to less productive locations leads to lower wages and a loss of population. The resulting loss of agglomeration economies reinforces this trend. Eventually, there is less demand for residential floor space so that a smaller horizontal zone is required to accommodate the population even conditional on a reduction in vertical space.

The effect of the height limit on residential rent is seemingly at odds with the intuition that supply constraints should increase housing cost. Yet, wages and rents in a frictionless open-city model must adjust to maintain the reservation utility level. If the height limit lowers the wage, the city must lose population until rents have fallen sufficiently to offset for the wage effect. For commercial rents, the effect of the height limit is more nuanced. For intermediate height limits, the supply effect of a more generous height limit offsets for the productivity effect, and commercial rents fall as the permittable height increases. It is the opposite for very stringent height limits. The loss of productivity along with the expansion in horizontal space, facilitated by lower competition for space from fewer residents, dominates the reduction in available vertical space, and rents fall as the permittable height decreases.

7.2. Welfare implications

In our open-city model, utility and profits are anchored to exogenous reservation levels. One city's gain from mobile workers is another city's loss. Hence, we focus on the immobile factor, land, to evaluate welfare. The bottom-right panel of Fig. 17 illustrates how the aggregate land rent over all uses (commercial, residential, agricultural) within the horizontal area of the unregulated city ($x \in (-x_1, x_1)$) changes as the city introduces a height limit. For a more intuitive interpretation, we normalize the change in land rent by the GDP ($g = \frac{yN}{\alpha^C}$) of the unregulated city.

To lend some further intuition to the counterfactuals, we note that our simulated city shares similarities with Houston, TX. Houston has a population of about two million, the tallest building (JPMorgan Chase Tower) has 75 floors, and the city is famous for being the only major city in the U.S. without formal zoning. Introducing a 10-floor height cap into such a city would reduce the aggregate land rent to the equivalent of 5.8% of the city's GDP. Given a 2012 GDP in Houston of about $400 billion, we can monetize the effect of a hypothetical height cap on annual land rent at $23 billion. Given a 2012 population of Houston of 2.1 million, this corresponds to about $11 thousand per capita and year. At a 5%-capitalization rate, the effect on land value would exceed $450 billion in total and $210 thousand in per-capita terms (see Online Appendix Section H.3).

The above thought experiment suggests that height caps can have very substantial welfare consequences. In our open-city model, we measure the incidence on the immobile factor: land. The alternative is to use a closed-city model in which workers are immobile and regulation leads to a reduction in worker welfare owing to greater commuting and housing cost.[26] Indeed, the public debate is often more concerned about the incidence of housing supply constraints on workers than on landlords. Also, much of the economics literature concerned with the measurement of the stringency of height regulations alludes to the closed-city case to derive predictions for housing affordability (Glaeser et al., 2005; Brueckner et al., 2017; Brueckner and Singh, 2020; Jedwab et al., 2020).

Whether to use the open-city or closed-city framework to study the effects of a height limit is a debatable issue. While evidence points to significant migration costs (Koşar et al., 2021), perfect mobility is a commonly accepted assumption of the canonical spatial equilibrium framework (Roback, 1982). Even if workers are imperfectly mobile, one takeaway from Fig. 17 is that allowing for taller buildings in an expensive city may be an imperfect solution to an affordability problem since it might only lead to more in-migration, higher wages, and correspondingly higher rents.[27] Hence, a relaxation of height restrictions will lead to a larger improvement of housing affordability if it is applied to more (if not all) cities within the relevant spatial system.

While the distortionary effects of height caps have been the primary concern of the related economics literature, it is important to acknowledge that height restrictions are implemented for a reason. Planners typically refer to a host of negative externalities associated with tall buildings, such as congestion or shadowing (see Online Appendix Section H.1).[28] Of course, the optimal building height will be smaller than the profit-maximizing height in the presence of negative externalities, rendering welfare effects of a height limit as theoretically ambiguous. A tentative conclusion from the counterfactuals in Fig. 17 and the literature reviewed in Online Appendix Section H.4 is that these negative externalities will have to be large to justify relatively strict height caps.[29]

8. Conclusion

One high-level conclusion of our synthesis is that the literature engaging with the economics of tall buildings is still at an early stage. We discuss the potential future research throughout Online Appendix Sections B to H. Here, we offer an admittedly subjective selection of three priority areas for research into the vertical dimension of cities.

First, there is a long way to go in understanding how the positive and negative externalities associated with tall buildings play out on balance. Skyscrapers may not just accommodate productive workers in productive locations but may also facilitate gains in productivity, ceteris paribus, if the vertical cost of interaction is sufficiently low. If well-designed skyscrapers put neighbourhoods or cities on the map, they may attract businesses and tourism. At this same time, skyscrapers may generate external costs in the form of shadows, congestion, or the loss of a coherent historic fabric.

Second, there is scope for exploring the costs of and returns to height more fully. The costs of and returns to height appear to be non-linear, and there are likely threshold effects that could be explored with richer data sets. Heterogeneity in the cost of height across uses and the value of the height amenities across users has implications for the horizontal pattern of land use and sorting are, thus, worth being incorporated into quantitative urban models. The net cost of height is one of the main congestive forces and a better understanding of how it evolves over time is helpful with respect to rationalizing changes in the spatial structure in the past as well as anticipating the evolution of cities in the future.

Third, there are forces outside the canonical competitive equilibrium framework that shape the urban height profile and deserve more attention. As an example, disentangling the effects of height competition from fundamentals that would justify extreme building heights remains an empirical challenge. Progress on this front will be essential for economically rationalizing skyscrapers such as the Empire State Building or the Burj Khalifa, which dominate height rankings for decades and exemplify the frontier of technological innovation and human ambition.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.jue.2021.103419

CRediT authorship contribution statement

Gabriel M. Ahlfeldt: Conceptualization, Formal analysis, Writing – original draft. **Jason Barr:** Conceptualization, Formal analysis, Writing – original draft.

References

Ahlfeldt, G.M., Albers, T., Behrens, K., 2020. Prime locations. CEPR Discuss. Paper 15470.
Ahlfeldt, G.M., Barr, J., 2020. Viewing urban spatial history from tall buildings. Reg. Sci. Urban Econ. 103618. doi:10.1016/j.regsciurbeco.2020.103618.
Ahlfeldt, G.M., McMillen, D.P., 2014. Land values in chicago, 1913–2010: A City's spatial history revealed. Land Lines (April).
Ahlfeldt, G.M., McMillen, D.P., 2018. Tall buildings and land values: height and construction cost elasticities in chicago, 1870–2010. Rev. Econ. Stat. 100 (5), 861–875.
Ahlfeldt, G.M., Pietrostefani, E., 2019. The economic effects of density: a synthesis. J. Urban Econ. 111 (February), 93–107. doi:10.1016/j.jue.2019.04.006.
Ahlfeldt, G.M., Redding, S.J., Sturm, D.M., Wolf, N., 2015. The economics of density: evidence from the berlin wall. Econometrica 83 (6). doi:10.3982/ECTA10876.
Ahlfeldt, G.M., Wendland, N., 2011. Fifty years of urban accessibility: the impact of the urban railway network on the land gradient in berlin 1890–1936. Reg. Sci. Urban Econ. 41 (2). doi:10.1016/j.regsciurbeco.2010.10.001.
Albouy, D., Arango, M., Shin, M., 2020. Crowning the metropolis: skylines, land values, and urban population. Working paper.
Alonso, W., 1964. Location and land use. Harvard Univ. Press, Cambridge, MA.
Autor, D.H., 2019. Work of the past, work of the future. AEA Paper. Proc. 109, 1–32. doi:10.1257/pandp.20191110.
Barr, J., 2010. Skyscrapers and the skyline: manhattan, 1895–2004. Real Estate Econ. 38 (3), 567–597.
Barr, J., 2013. Skyscrapers and skylines: new york and chicago, 1885–2007. J. Reg. Sci. 53 (3). doi:10.1111/jors.12017.
Barr, J., 2016. Building the skyline: The birth and growth of Manhattan's skyscrapers. Oxford University Press.
Barr, J., Luo, J., 2020. Growing skylines: the economic determinants of skyscrapers in china. J. Real Estate Financ. Econ..
Barr, J., Mizrach, B., Mundra, K., 2015. Skyscraper height and the business cycle: separating myth from reality. Appl. Econ. 47 (2). doi:10.1080/00036846.2014.967380.
Barr, J., Smith, F., Kulkarni, S., 2018. What'S manhattan worth? a land values index from 1950 to 2014. Reg. Sci. Urban Econ. 70. doi:10.1016/j.regsciurbeco.2018.02.003.

[26] Bertaud and Brueckner (2005) formally derive the intuitive predictions that binding height limits force cities to expand horizontally if the city population is fixed. Exploiting height limits in Indian cities, Brueckner and Sridhar (2012) empirically confirm the central prediction that there is a negative relationship between the height limit and the geographic area of city. Their estimates imply that relaxing height limits would increase welfare substantially.

[27] With skill-biased returns to agglomeration, the distributional effects could be regressive (Ahlfeldt and Pietrostefani, 2019).

[28] Indeed, some cities such as London differentiate with respect to the reputation of the architect when enforcing height limits to avoid negative visual externalities (Cheshire and Dericks, 2020).

[29] In horizontal settings, evidence suggests that the net effect of land use regulation is negative (Cheshire and Sheppard, 2002; Turner et al., 2014).

Barr, J., Tassier, T., Trendafilov, R., 2011. Depth to bedrock and the formation of the manhattan skyline, 1890–1915. J. Econ. Hist. 71 (4), 1060–1077.

Bartik, T., 1991. Who benefits from state and local economic development policies? doi:10.17848/9780585223940. Kalamazoo, MI

Bascomb, N., 2004. Higher: A Historic race to the sky and the making of a city. Crown, New York.

Baum-Snow, N., 2007. Did highways cause suburbanization? Q. J. Econ. 122 (2), 775–805. doi:10.1162/qjec.122.2.775.

Bercea, B., Ekelund, R.B., Tollison, R.D., 2005. Cathedral building as an entry-Deterring device. Kyklos 58 (4), 453–465. doi:10.1111/j.0023-5962.2005.00297.x.

Berger, J.S., 1967. Determination of the economic height of high-rise buildings. University of California, Los Angeles. Ph.D. Dissertation.

Bertaud, A., Brueckner, J.K., 2005. Analyzing building-height restrictions: predicted impacts and welfare costs. Reg. Sci. Urban Econ. 35 (2), 109–125.

Brooks, L., Lutz, B., 2016. From Today's city to Tomorrow's city: an empirical investigation of urban land assembly. Am. Econ. J.: Econ. Policy 8 (3), 69–105. doi:10.1257/pol.20130399.

Brueckner, J.K., 1987. The structure of urban equilibria: a unified treatment of the muth-mills model. Handbook Region. Urban Econ. 2, 821–845. https://econpapers.repec.org/bookchap/eeeregchp/2-20.htm

Brueckner, J.K., 2000. Urban Growth Models with Durable Housing: An Overview. In: Huriot, J.-M., Jacques-François, T. (Eds.), Economics of Cities: Theoretical Perspectives. Cambridge University Press Cambridge, pp. 263–289.

Brueckner, J.K., Fu, S., Gu, Y., Zhang, J., 2017. Measuring the stringency of land use regulation: the case of China's building height limits. Rev. Econ. Stat. 99 (4), 663–677.

Brueckner, J.K., Singh, R., 2020. Stringency of land-use regulation: building heights in US cities. J. Urban Econ. 116.

Brueckner, J.K., Sridhar, K.S., 2012. Measuring welfare gains from relaxation of land-use restrictions: the case of India's building-height limits. Reg. Sci. Urban Econ. 42 (6), 1061–1067.

Cheshire, P., Sheppard, S., 2002. The welfare economics of land use planning. J. Urban Econ. 52 (2), 242–269. doi:10.1016/S0094-1190(02)00003-7.

Cheshire, P.C., Dericks, G.H., 2020. Trophy architects and design as rent-seeking: quantifying deadweight losses in a tightly regulated office market. Economica.

Clark, W.C., Kingston, J.L., 1930. The skyscraper: Study in the economic height of modern office buildings. American Institute of Steel.

Combes, P.-P., Duranton, G., Gobillon, L., 2011. The identification of agglomeration economies. J. Econ. Geogr. 11 (2), 253–266.

Combes, P.-P., Duranton, G., Gobillon, L., 2019. The costs of agglomeration: house and land prices in french cities. Rev. Econ. Stud. 86 (4), 1556–1589. doi:10.1093/restud/rdy063.

Combes, P.-P., Duranton, G., Gobillon, L., 2021. The production function for housing: evidence from france. J. Polit. Econ..

Combes, P.-P., Gobillon, L., 2015. The Empirics of Agglomeration Economies. In: Duranton, G., Henderson, J.V., Strange, W.C. (Eds.), Handbook of Regional and Urban Economics, Vol. 5. Elsevier, pp. 247–348.

Danton, J., Himbert, A., 2018. Residential vertical rent curves. J. Urban Econ. 107, 89–100. doi:10.1016/j.jue.2018.08.002.

Duranton, G., Puga, D., 2015. Urban Land Use. In: Duranton, G., Henderson, J.V., Strange, W.C. (Eds.), Handbook of Regional and Urban Economics, Vol. 5. Elsevier, pp. 467–560. https://econpapers.repec.org/RePEc:eee:regchp:5-467

Duranton, G., Turner, M.A., 2012. Urban growth and transportation. Rev. Econ. Stud. 79 (4), 1407–1440.

Environment, D.o.t., 1971. Local authority offices: Areas and costs. DOE, London.

Fujita, M., Ogawa, H., 1982. Multiple equilibria and structural transition of non-monocentric urban configurations. Reg. Sci. Urban Econ. 12 (2), 161–196. doi:10.1016/0166-0462(82)90031-X.

Glaeser, E.L., Gyourko, J., Saks, R., 2005. Why is manhattan so expensive? regulation and the rise in housing prices. J. Law Econ. 48 (October), 331–369. doi:10.1086/429979.

Glaeser, E.L., Kallal, H.D., Scheinkman, J.A., Shleifer, A., 1992. Growth in cities. J. Polit. Econ. 100 (6), 1126–1152. http://www.jstor.org/stable/2138829

Glaeser, E.L., Kolko, J., Saiz, A., 2001. Consumer city. J. Econ. Geogr. 1 (1), 27–50. doi:10.1093/jeg/1.1.27. http://joeg.oxfordjournals.org/content/1/1/27.abstract

Helsley, R.W., Strange, W.C., 2008. A game-theoretic analysis of skyscrapers. J. Urban Econ. 64 (1), 49–64. doi:10.1016/J.JUE.2007.08.004. http://www.sciencedirect.com/science/article/pii/S0094119007000952

Henderson, V.J., Kuncoro, A., Turner, M.A., 1995. Industrial development in cities. J. Polit. Econ. 103 (5), 1067–1090. http://www.jstor.org/stable/2138755

Hoyt, H., 1933. 100 Years of land values in chicago: The relationship of the growth of chicago to the rise in its land values, 1830–1933. University of Chicago Press.

Hsieh, C.-T., Moretti, E., 2019. Housing constraints and spatial misallocation. Am. Econ. J.: Macroeconomics 11 (2), 1–39. doi:10.1257/mac.20170388.

Jedwab, R., Barr, J., Brueckner, J. K., 2020. Cities without Skylines: Worldwide Building-height Gaps and their Implications.

Kim, H.I., 2004. Space efficiency in mixed-use high-rise building. Illinois Institute of Technology Ph.D. Dissertation.

Kingwell, M., 2006. Nearest thing to heaven: The empire state building and american dreams. Yale University Press, New Haven, Connecticut.

Koşar, G., Ransom, T., van der, K.W., 2021. Understanding migration aversion using elicited counterfactual choice probabilities. J. Econom. forthcomin.

Lee, J.-S., Lee, H.-S., Park, M.-S., 2011. Schematic cost estimating model for super tall buildings using a high-rise premium ratio. Can. J. Civ. Eng. 38 (5), 530–545. doi:10.1139/L11-025.

Liu, C., Rosenthal, S.S., Strange, W.C., 2018. The vertical city: rent gradients and spatial structure. J. Urban Econ. 106, 101–122. doi:10.1016/j.jue.2018.04.001.

Liu, C., Rosenthal, S.S., Strange, W.C., 2020. Employment density and agglomeration economies in tall buildings. Reg. Sci. Urban Econ. 84.

Lucas, R.E., Rossi-Hansberg, E., 2002. On the internal structure of cities author. Econometrica 70 (4), 1445–1476.

Mills, E.S., 1967. An aggregative model of resource allocation in a metropolitan area. Am. Econ. Rev. 57 (2), 197–210. http://www.jstor.org/stable/1821621

Moretti, E., 2012. The new geography of jobs. Houghton Mifflin Harcourt.

Muth, R.F., 1969. Cities and housing. Chicago University Press, Chicago.

Redding, S.J., Turner, M.A., 2015. Transportation Costs and the Spatial Organization of Economic Activity. Handbook of Regional and Urban Economics, Vol. 5 doi:10.1016/B978-0-444-59531-7.00020-X.

Roback, J., 1982. Wages, rents, and the quality of life. J. Polit. Econ. 90, 1257–1278.

Rosenthal, S.S., Strange, W.C., 2008. The attenuation of human capital spillovers. J. Urban Econ. 64 (2), 373–389. doi:10.1016/j.jue.2008.02.006. http://www.sciencedirect.com/science/article/pii/S0094119008000193

Rosenthal, S.S., Strange, W.C., Urrego, J.A., 2021. JUE Insight: are city centers losing their appeal? commercial real estate, urban spatial structure, and COVID-19. J. Urban Econ. forthcomin.

Saiz, A., 2010. The geographic determinants of housing supply. Q. J. Econ. 125, 1253–1296. doi:10.2307/27867510.

Sev, A., Özgen, A., 2009. Space efficiency in high-rise office buildings. Metu Jfa 26 (2), 69–89.

Smith, B., Dholakia, M., Heseltine, R., 2014. Cost model: tall buildings. Building April 15.

Spengler, E.H., 1930. Land values in new york in relation to transit facilities. Columbia University Press.

Strange, W.C., 1995. Information, holdouts, and land assembly. J. Urban Econ. 38 (3), 317–332.

Tan, W., 1999. Construction cost and building height. Construct. Manag. Econ. 17 (2), 129–132. doi:10.1080/014461999371628.

von Thuenen, J.H., 1826. Der isoli[e]rte staat in beziehung auf landwirtschaft und nationalökonomie, oder untersuchungen über den einfluß, den die getreidepreise, der reichthum des bodens und die abgaben auf den ackerbau ausüben. Verlag von Wiegandt,, Hamburg.

Titman, S., 1985. Urban land prices under uncertainty. Am. Econ. Rev. 75 (3), 505–514. http://www.jstor.org/stable/1814815

Turner, M.A., Haughwout, A., van der Klaauw, W., 2014. Land use regulation and welfare. Econometrica 82 (4), 1341–1403. doi:10.3982/ECTA9823.

Watts, S., Kalita, N., Maclean, M., 2007. The economics of super-tall towers. Struct. Design Tall Spec. Build. 16 (4), 457–470.

Williams, J.T., 1991. Real estate development as an option. J. Real Estate Finance Econ. 4 (2), 191–208.